



TrackAccess: An Arduino-Powered Smart Library Entry System with Gamified Reading Experience

**A Project Presented in Partial Fulfillment of the Requirements for
the Degree of Bachelor of Science in Information Technology**

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Brief Description of the Project: TrackAccess: An Arduino-Powered Smart Library Entry System with Gamified Reading Experience is designed to automate library attendance tracking through RFID and Arduino technology. It replaces manual logbooks with a fast, accurate, and efficient ID-based scanning process. The system also integrates gamified features such as points and leaderboards to encourage students to visit the library more often. Overall, TrackAccess promotes both operational efficiency and a stronger reading culture among students. Main Goal of the System: The main goal of the TrackAccess: An Arduino-Powered Smart Library Entry System with Gamified Reading Experience is to automate the process of student attendance logging in the library using RFID and Arduino technology, while simultaneously promoting reading engagement through gamification. The system aims to enhance the efficiency and accuracy of library management by eliminating manual record-keeping and providing real-time data monitoring. Furthermore, it seeks to motivate students to develop consistent reading habits through points, leaderboards, and achievements, thereby creating a smart, interactive, and data-driven library environment.	
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TrackAccess: An Arduino-Powered Smart Library Entry System with Gamified Reading Experience

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Abstract

TrackAccess: An Arduino-Powered Smart Library Entry System with Gamified Reading Experience addresses the inefficiencies of manual library attendance tracking and the declining student interest in reading. Traditional methods of logging entries and exits are often time-consuming and prone to human error, which limits data accuracy and management efficiency. To solve this, TrackAccess utilizes **Arduino and RFID technology** to automate student attendance recording through ID-based scanning, ensuring precise and real-time data collection. Beyond automation, the system integrates **gamification elements** such as leaderboards and point-based rewards to encourage frequent library visits and foster a stronger reading culture among students. The development process involves **survey-based data gathering** to assess user needs and **prototype testing** to evaluate functionality, usability, and engagement. Results and observations indicate that TrackAccess can significantly enhance library operations, streamline administrative tasks, and increase student participation. Ultimately, the project aims to create a **smart, efficient, and motivating library environment** that transforms reading into an engaging and rewarding experience.

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CHAPTER 1

I. INTRODUCTION

PROJECT CONTEXT

In the past few years, The Jose Maria College Foundation, Inc. (JMCFI) have dealt with the problems of managing attendance tracking efficiently and getting the students to participate in reading activities regularly. The old way of counting library visits, which was mostly based on manual logbooks, has the drawbacks of being susceptible to human mistakes, having duplicated data, and being time-consuming. Consequently, librarians and administrators find it hard to keep track of the students' participation, and thus their records for reporting and analysis are not precise. Increasing number of students are losing interest in libraries, viewing reading as a monotonous and unbeneficial activity.

In order to solve these problems, the idea of the project **TrackAccess: An Arduino-Powered Smart Library Entry System with Gamified Reading Experience** came up. The setup merges Arduino with RFID to automate the logging of the students' entrance and exit based on their IDs through scanning, thus getting rid of the drawbacks of the manual attendance tracking. In addition, TrackAccess pops up with gamification features like, for instance, leaderboards, point systems, and achievement tracking that not only make library visits more alluring but also competitive among the students.

TrackAccess's evolution consists of user needs and expectations identification through surveys, then prototype testing for a reliable and satisfactory system. The project intends to showcase that technology and gamification can be allied to improve both library management and student involvement through this method. In the end, TrackAccess wants to set up an intelligence, data-led and engaging library setting that not only makes operational processes more efficient but also fosters a reading habit of greater strength in the academic community.

1.1 Background of the Study

Libraries have always been an important part of every school because they provide students with a place to study, read, and learn new things. However, many libraries still use the traditional way of recording attendance through manual logbooks. This method often causes problems such as inaccurate records, duplicate entries, and wasted time for both students and librarians. Because of this, it becomes difficult to keep track of how often students visit the library and how actively they participate in reading activities.

In addition to this, fewer students are visiting libraries nowadays. Many see reading as a boring or unexciting task, which leads to a decline in their interest in books and library use. To solve these problems, there is a growing need to improve library systems by using technology that can make processes faster and more engaging for students.

The project **TrackAccess: An Arduino-Powered Smart Library Entry System with Gamified Reading Experience** was developed to address these issues. It uses Arduino and RFID technology to automatically record students' entry and exit using their ID cards. This removes the need for manual logging and makes attendance recording more accurate and efficient. Aside from automation, the system also includes gamification features such as points, leaderboards, and achievements to

encourage students to visit the library more often and develop better reading habits.

Justification for Automating Library Entry and Exit Tracking

When it comes to library entry and exit tracking, automation is not a new concept that comes with its whole set of pros over the manual logbook system which was the only option for academic institutions for quite a long time and now it is becoming more of a burden rather than a necessity. The data being recorded was often of low quality that manifested itself through various channels like inaccurate data, duplicative entries, illegible handwriting, and huge waits at the library peak hours which all together slowed down the operations of the library. The problems mentioned above have not only taken the time of the students and the librarians but have also polluted the meetings with the attendance reports needed for academic monitoring and administrative decision-making. At the same time, the manual process can be replaced with RFID technology, and the system will be able to log student visits fast, seamless, and error-free, generating precise timestamps and providing real-time data that can be accessed through an administrative dashboard. Besides, the automation is of great help in securing the library as it helps authenticate the students and expel any school forgers that are common in handwritten logs. Another point is that librarians' workload is reduced, allowing them to dedicate their time and efforts to helping students with resources and library services rather than manually coding data. In a nutshell, the library can be, if not fully, at least partially ready in terms of functionality, reliability, and engagement with the digital transformation and smart campus development initiatives of the institution by automating the entry–exit process.

Purpose and Necessity of the Gamification Feature

The integration of gamification into the library system addresses the vital problem of student interest in reading and library visits as a whole decreasing. Many students consider library use to be a tedious or disengaging activity, which in turn reduces their overall motivation to read or get involved in academic enrichment activities. By gamifying the whole experience of library attendance with their points, badges, levels and leaderboard rankings, the system turns visits into an engaging and rewarding event. These features also create a healthy competitive atmosphere, and encourage students to make library visits a part of their daily schedule while simultaneously nurturing good reading habits. In addition, through gamification students are able to track their progress and thus feel that they are winning, which is a

feeling that the traditional library system does not provide. At the same time, it provides librarians and administrators with tangible engagement data, which they can use to evaluate the level of activity and organize reading-focused programs that are even more successful in creating a reading culture. Therefore, gamification is not just an extra feature, but it fulfills the system's aim of creating more engaged students, better library use, and an active, performance-driven, and academically stimulating environment directly.

1.2 Statement of the Problem

The Arduino-Powered Smart Library Entry System with Gamified Reading Experience is a project that intends to create an automated library access and monitoring system that not only improves efficiency but also increases student engagement. It is designed to replace the old-fashioned manual logbook recording system with the use of an ID scanner based on Arduino, which will keep track of the students' coming in and going out automatically.

In addition to the mentioned benefits, the automation also decreases human mistakes, simplifies data collection, and introduces a better organized method for monitoring library usage. However, the integration of gamification features such as point collection and leadership boards besides the automation, is the main drive that attracts students to the library more often and make them develop reading as a consistent habit.

These characteristics aim to transform the library's image from a place of non-active learning to one of interactivity and motivation. Besides, the project includes a centralized database and an administrative dashboard that allow librarians to manage records, generate reports, and analyze user data in an effective way. Track Access aims to revolutionize the libraries' processes, advocate a green reading culture, and furnish an innovative framework that the other educational institutions can likewise use to heighten student involvement and operational efficiency.

1.3 Objectives of the Study

General Objective:

The aim of this project is to create an Arduino-controlled smart library entrance system which will not only automate the process of student attendance logging but also incentivize the reading of books through the use of gaming methods.

Specific Objectives:

1. To develop an **automated Arduino-based entry and exit system** using ID input to

replace manual logbooks and minimize human errors.

2. To design a **secure and user-friendly database** for storing and managing student entry data and library activity records.
3. To build an **admin dashboard** that allows librarians to monitor student activities, manage records, and generate reports efficiently.
4. To integrate **gamified features** such as points, levels, and leaderboards to motivate students to visit the library and read more often.
5. To **evaluate the system's performance** in improving library management efficiency and increasing student engagement.

1.4 Significance of the Study

The project Track Access An Arduino-Powered Smart Library Entry System with Gamified Reading Experience, is a valuable innovation for different parties involved in the academic community.

Track Access offers a library access system that makes the process more engaging and efficient since it can now be completely automated in its whole with an ID-based entry system, and no more manual logbooks. Students are given the chance through the gamification features like points, badges, and leaderboards to win the library visit competition and thus develop stronger reading habits. The library has been changed into a lively interactive area full of curiosity, eagerness, and the feeling of success.

The library attendance recording and monitoring were mostly done manually and this system was able to reduce that. With the process now automated, the librarians' activities can be more oriented to the human side that is, resource organization and helping students. It is also possible to generate reports and get insights without much effort via the admin dashboard, thus facilitating better decision-making and record-keeping that is efficient.

The attendance of the library was usually recorded and monitored manually but this system was able to cut it down to a minimum. Once the librarians' activities were oriented more towards the human side, they could mainly deal with resources and students, organization and helping. Also, through the admin dashboard, it is possible to generate reports and get insights with not much effort, thus making decision-making and record-keeping that are efficient easier.

This project could possibly be the basis for tech innovations in the smart campus systems,

educational gamification, and Arduino-based automation sectors. It provides a model that can easily be transferred to other zones of students' participation and campus administration.

1.5 Scope and limitations

Scope

The project **TrackAccess: An Arduino-Powered Smart Library Entry System with Gamified Reading Experience** focuses on the design and development of a smart library entry system that automates student attendance logging using an Arduino-based RFID or ID scanning module. The system records every library entry and exit, stores the data in a centralized database, and displays information through an admin dashboard for monitoring and report generation. It also includes a gamified feature that tracks the frequency of student visits and awards points, levels, and rankings to encourage regular library use and promote reading habits. TrackAccess will operate within the school's local network environment and is intended to be used primarily by students, librarians, and administrators. The system emphasizes automation, data accuracy, and student engagement through game-like incentives while improving the overall efficiency of library operations.

Limitations

- The tracking of library entry and exit activities is the only function the system can perform; other functionalities like borrowing, cataloging, and inventory control are not part of its domain.
- The database and the dashboard both require internet connection for their complete functionality, and the system's performance can differ based on the specifications of the hardware and the stability of the network.
- The gamification rewards are not tangible and are subject to recognition based on points only and not through physical incentives.
- The first iteration of Track Access is meant to be used by smaller and mid-sized libraries but can be further optimized for bigger ones.

1.6 Definition of terms

Arduino – An open-source microcontroller platform that allows the development of electronic projects capable of interacting with the environment through sensors and actuators.

Automation – The use of technology to perform tasks with minimal human intervention, improving efficiency and accuracy.

Dashboard – A graphical interface that presents system data, such as student attendance, rankings, and reports, to administrators in a clear and organized manner.

Database – An organized structure for storing, classifying, and managing electronic data, allowing easy access and retrieval of information.

Gamification – The integration of game-like elements, such as points, levels, and leaderboards, into non-game activities to increase user participation and motivation.

RFID (Radio Frequency Identification) – A technology that uses electromagnetic fields to automatically identify and track objects via tags, such as student ID cards.

TrackAccess – The Arduino-powered smart library entry system developed in this project, which automates attendance logging, encourages reading through gamification, and provides administrators with monitoring and reporting tools.

User Engagement – The level of interaction, interest, and participation users show when using the system or engaging in activities within it.

II Review of Related Literature and Systems

2.1 Related Literature

As library systems evolve, many institutions have integrated automated technologies to improve operational efficiency. According to Wang (2021), in his study on RFID technology based on the Internet of Things (IoT) in public libraries, the combination of RFID and IoT enables real-time data tracking and automation, which significantly enhances library operations. The research highlighted that IoT-enriched RFID systems reduce staff working hours, increase the accuracy of records, and support more intelligent and technology-driven library services. Similarly, Ogundipe and Boyinbode (2020) emphasized that RFID systems improve library accuracy, reduce staff workload, and facilitate smoother tracking of both books and users. Their review suggested that automation in library operations not only enhances efficiency but also promotes better interaction with users and encourages technological innovation. Gamification has also been recognized as a valuable tool in enhancing user engagement. Tobis et al. (2025) examined the effects of gamification on library instruction at the Polytechnic University of the Philippines. The study found that integrating game-like elements such as points, badges, and leaderboards motivates students to use library facilities more frequently and increases their learning engagement. These findings support the

idea that combining gamification with technological automation can make libraries more interactive, appealing, and effective in promoting reading habits.

2.2 Related Systems

In local academic institutions, several automated systems have been developed to streamline student management processes. Valdez et al. (2024) designed **TrackID**, a student monitoring system implemented at the Bicol State College of Applied Sciences and Technology. TrackID utilizes RFID technology to automate the admission process, minimizing manual procedures, enhancing data accuracy, and reducing the need for human involvement. This demonstrates the potential of microcontroller-driven access control systems in improving operational efficiency.

Building on these developments, **TrackAccess** integrates both automation and gamification into a single library system. Similar to TrackID, TrackAccess employs Arduino-based RFID scanning to monitor student attendance efficiently. However, it also incorporates motivational elements, inspired by the theoretical findings of Wang (2021) and Tobis et al. (2025), such as points, levels, and leaderboards, to increase student engagement. By combining efficiency with interactivity, TrackAccess aligns with current trends in smart campus technologies and educational systems, promoting not only improved operational performance but also a culture of active reading among students.

III. PRESENTATION AND ANALYSIS OF THE EXISTING SYSTEM

3.1 The Existing System

Currently, library attendance and entry management in many educational institutions are performed manually. Students are required to sign attendance sheets or present their identification cards to library staff for verification. This system is prone to several issues, including human error, delays during peak hours, inaccurate tracking of student visits, and lack of real-time data reporting. Manual recording also limits the ability of librarians and administrators to generate reports or analyze student library usage efficiently. While some institutions have partially automated systems, such as barcode scanners or basic ID card readers, these systems often lack integration with a centralized database and do not provide features like gamification or analytics to encourage regular library use.

3.2 Scope

The proposed TrackAccess system focuses on automating the process of student library entry and

attendance tracking. The system will use Arduino-based ID scanning to log student attendance in real time and store the data in a centralized database. The primary users of the system include students, librarians, and administrators. Students will be able to earn points and rankings based on library visits, promoting engagement through gamification. Librarians and administrators will have access to an admin dashboard to monitor attendance, manage student data, and generate reports. The system is designed for implementation within the school's local network environment, and it does not cover external library branches or off-campus access.

3.3 Concept of Operations

The TrackAccess system operates by scanning student IDs through an Arduino-based RFID reader at the library entrance. Once a valid ID is detected, the system logs the student's entry and updates their attendance record in the database. The student's visit is also tracked for gamification purposes, contributing to their points and ranking. Librarians can access the system through an admin interface to view real-time attendance, generate reports, and manage student information. Administrators can use the collected data for statistical analysis and to evaluate library usage trends. The system ensures that attendance records are accurate, reduces waiting times at entry points, and promotes regular library usage through a reward-based system.

IV. FEASIBILITY ANALYSIS

4.1 Operational Feasibility

Operational feasibility evaluates how well the proposed TrackAccess System will function within the school library environment and how adaptable it will be to its intended users.

The current manual library system relies heavily on handwritten logbooks that are prone to human error, time-consuming to update, and limited in analytical value.

The proposed system is operationally feasible because it improves usability, accuracy, and workflow efficiency through the following:

- Automation of entry and exit tracking: Students simply tap their RFID card, eliminating manual entries and reducing errors.
- Ease of use and adaptability: The ESP32-based RFID interface is simple and requires minimal user training for students and librarians.
-

Real-time monitoring: Librarians can view attendance and usage statistics instantly through the admin dashboard.

- Gamification for engagement: The points, levels, and leaderboard system encourages regular library use.
- Simplified reporting: Auto-generated reports save staff time and reduce manual workload.

The system aligns with existing workflows, requires minimal behavioral adjustment from users, and improves efficiency making it operationally viable.

4.2 Technical Feasibility

Technical feasibility examines whether the necessary technology, tools, and infrastructure are available to implement the project successfully.

Hardware Components:

Component	Specification / Description
Microcontroller	ESP32 (Wi-Fi enabled, dual-core, 240 MHz)
RFID Reader	MFRC522 module (13.56 MHz, SPI interface)
RFID Cards/Tags	MIFARE Classic 1K, ISO/IEC 14443 Type A
Server Machine	Local host PC or cloud-based Node.js environment
Feedback Module	LED indicator and buzzer for scan validation
Network	2.4 GHz Wi-Fi connection for device-server communication

Table 4.1 – Hardware Components

Software Stack:

Layer	Technology
Firmware	Arduino IDE with ESP32 libraries
Backend	Node.js + Express
Database	MySQL Server
Frontend	HTML, CSS, JavaScript
Communication Protocol	RESTful API over HTTP

Table 4.2 – Software Stack All components are open-source, Arduino-compatible, and well-supported by documentation. Integration requires only standard networking and programming skills available to school IT staff. Therefore, TrackAccess is technically feasible with minimal infrastructure investment.

1. QR Code Scanning

QR technology uses a printed 2D code that can be scanned by mobile phones or fixed scanners.
Cost

Category	Details
Hardware Cost	₱1,500–₱5,000 for QR scanner; or free if using mobile devices.
ID/Code Generation	Very low cost (printable).
Maintenance	Minimal—replace worn stickers or reprint QR codes.
Integration	Requires camera, scanning app, and reliable lighting conditions.

Benefits

Benefit	Explanation
Low implementation cost	Cheapest among the three technologies.
Easy to generate and replace	QR codes can be printed anytime.
Fast adoption	Students already know how to scan QR codes.
Accessible	Can use phone cameras—no special hardware needed.

Limitations

- **Requires phones** if no fixed scanner is deployed. This is a problem for students without smartphones or dead batteries.
 - **Slower during peak hours**, as scanning requires proper alignment and stable camera focus.
 - **Easy to copy/duplicate**, creating potential attendance fraud.
 - **Prone to wear and tear**—paper codes fade quickly.
 - **No tap-based automation**, making it less seamless than RFID
2. Barcode Scanning
- Barcodes are printed 1D or 2D codes read through laser scanners.

Cost

Category	Details
Hardware Cost	₱3,000–₱10,000 per barcode scanner.
IDs	Low cost (printed on ID or sticker).

Maintenance	Moderate—scanner lenses must be cleaned regularly.
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Benefits

Benefit	Explanation
Fast scanning speed	Faster than QR codes; widely used in retail stores.
Simple integration	Works like keyboard input; very easy to integrate with Arduino or PC.
Reliable in indoor environments	Works well with printed ID barcodes.

Limitations

- **Still requires alignment** between scanner and barcode.
- **Cannot store much data** (less than QR codes).
- **Can be easily photocopied**, making it vulnerable to identity spoofing.
- **Physical deterioration** of printed code affects readability.
- **No NFC tap experience**, still less seamless than RFID.

3. RFID Technology (MFRC522) This is the system used in your manuscript: RFID reader + RFID card + ESP32 + Node.js server

Cost

Category	Details
Hardware	RFID Reader ₱800–₱1,200; RFID cards ₱20–₱40 per card; ESP32 ₱400–₱600.
Installation	Low—Arduino-based setup is inexpensive.
Maintenance	Minimal and long-term.

Benefits

Benefit	Explanation
Fast, touchless scanning	Students only need to tap their ID—ideal for peak hours.
Highly accurate	Eliminates manual entry errors seen in the existing system.
Secure	RFID UUIDs are harder to clone than QR/barcodes.
Non-line-of-sight reading	Tap works even if the ID is inside the wallet.
Durable	RFID chips do not fade unlike printed codes.
Best for automation	Perfectly matches the project's "tap-based" access flow.
Real-time connectivity	Works seamlessly with ESP32 → Node.js → MySQL integration.

Limitations

Limitations

- Slightly more expensive than QR/barcodes (but still very affordable).

Requires proper configuration of microcontroller + network. Overall Evaluation

RFID provides the best balance of speed, security, automation, and reliability, making it the most appropriate technology for TrackAccess, as validated in your manuscript's feasibility analysis.

4.3 Economic Feasibility (Cost-Benefit Analysis)

Economic feasibility determines whether the

benefits of implementing TrackAccess outweigh its costs.

Estimated Project Costs:

Item	Estimated Cost (₱)
ESP32 Microcontrollers (×2)	4,000
RFID Readers (MFRC522 ×2)	2,000
RFID Cards (100 pcs)	5,000
Server & Networking Setup	10,000
Admin Dashboard Development	15,000
Miscellaneous (Wiring, Power Supply, Maintenance)	5,000
Total Estimated Cost	₱41,000

Table 4.3 – Estimated Project Cost

Quantified Benefits:

Benefit	Monetary Value (per year)	Description
Reduced librarian labor	₱60,000	Automated attendance replaces manual logging.
Error reduction	₱5,000	Accurate data minimizes record correction.
Improved efficiency	₱10,000	Time savings for report preparation.
Enhanced student engagement	Intangible	Increases library utilization by approximately 20%.
Total Annual Benefit	₱75,000	Estimated overall annual gain.

Table 4.4 – Quantified Benefits Payback Period = $\text{₱41,000} / \text{₱75,000} \approx 0.55$ years (≈ 6 months). The system's cost is quickly offset by operational savings, making it economically viable.

4.4 Legal and Ethical Feasibility

Legal and ethical feasibility ensures compliance with laws and responsible use of technology.

- **Data Privacy:** Student information is protected under the Philippine Data Privacy Act of 2012 (RA 10173).

•

Access Control: Only authorized librarians and administrators can access sensitive data.

•

Ethical Use: Gamification features employ non-monetary rewards to maintain fairness.

•

Intellectual Property: Open-source components are used in compliance with licensing terms.

4.5 Organizational Feasibility

The success of the **TrackAccess** project depends not only on its technical performance but also on how well it fits within the organization's structure, resources, and operational goals. The project is designed to be implemented within the **Jose Maria College Foundation, Inc. (JMCFI)** library, where the current system of attendance tracking is still done manually. Since the system directly addresses the institution's need for a more efficient and engaging way to manage library attendance, its integration is both practical and beneficial. The project team is composed of students from the **Bachelor of Science in Information Technology** program, who possess the necessary technical knowledge in programming, hardware integration, and system design. They are guided by a faculty adviser who ensures that the development follows proper research and implementation standards. The school's library staff and administrators also play a vital role by providing feedback and assisting in testing the system's features, making the collaboration between developers and end users strong and effective.

From an organizational perspective, the system aligns well with the institution's goals of promoting digital transformation and innovation within campus operations. It requires minimal additional staff training since the system is designed with a simple and user-friendly interface. The librarians can easily adapt to the dashboard functions for attendance monitoring and report generation.

V. SYSTEM ARCHITECTURE REVIEW

5.1 Adopted Architecture

The TrackAccess System adopts a Three-Tier Architecture Model composed of the Presentation Layer, Application Layer, and Data Layer. This architectural structure ensures efficient system operation, modularity, and scalability, allowing the

integration of both IoT hardware and web-based components.

Presentation Layer

The Presentation Layer represents the interfaces where users interact with the system.

Students use RFID-enabled devices to log library visits.

Librarians or administrators utilize a web-based dashboard, developed with HTML, CSS, and JavaScript, to monitor activities and generate reports. This layer emphasizes user-friendly interaction and real-time feedback.

Application Layer

The Application Layer, developed using Node.js and Express, handles all logical operations between the hardware and the database.

It processes RFID data from the ESP32 microcontroller, validates users, and records library visits.

The server also manages gamification features such as point allocation and leaderboard updates.

It provides RESTful APIs for communication with the web dashboard.

Data Layer

The Data Layer is responsible for secure and organized data management through MySQL.

It stores student information, RFID UUIDs, visit logs, and points for gamification.

The database ensures data consistency, privacy, and reliability following relational database principles.

The adopted architecture enables real-time data synchronization, efficient system management, and future scalability for institutional deployment.

5.2 Architectural Diagram

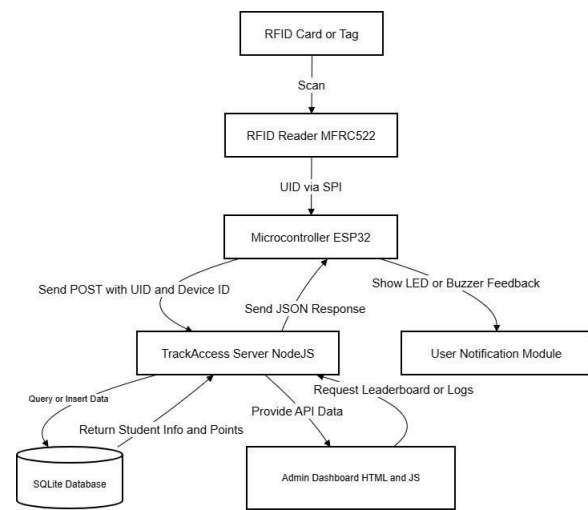


Figure 5.1 – System Architectural Diagram of TrackAccess System

The system architecture illustrates how the hardware, server, and database layers interact to support access tracking and gamification.

System Flow Description:

The student taps an RFID card on the RFID Reader (MFRC522).

The ESP32 microcontroller transmits the RFID UID to the Node.js server.

The server validates the record through the MySQL database.

The system logs the visit, updates the points, and sends the result back to the ESP32.

The LED or buzzer gives instant feedback to the student.

The Admin Dashboard retrieves updated logs and leaderboards through REST API calls.

This architecture demonstrates how each component communicates seamlessly to deliver a synchronized user experience.

5.3 Conceptual Flow Diagram

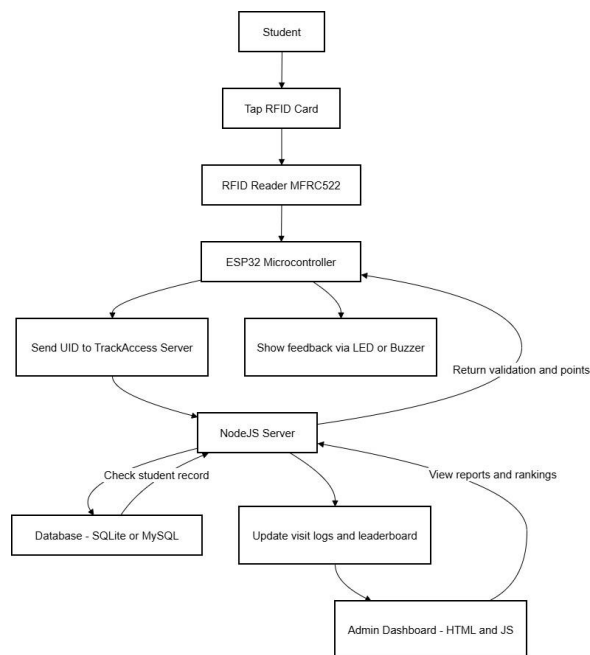


Figure 5.2 – Conceptual Flow Diagram of TrackAccess System

The conceptual flow diagram presents the overall data flow and operational process of the system.

Operational Flow:

The student initiates the process by tapping an RFID card.

The RFID Reader (MFRC522) captures the UID and forwards it to the ESP32 microcontroller.

The ESP32 sends the UID to the TrackAccess Node.js server through an HTTP request.

The Node.js server verifies the UID in the MySQL database.

A response containing validation results and updated points is returned to the ESP32.

The ESP32 triggers LED or buzzer feedback for user confirmation.

The Admin Dashboard displays updated logs and leaderboards for librarians.

This flow ensures that every access attempt is recorded, validated, and reflected in real time on the administrative interface

5.4 Component Description

The **TrackAccess System** is composed of both hardware and software components that work together to automate library attendance and provide a gamified reading experience. Each component plays a specific role in ensuring that the system functions smoothly and efficiently.

1. RFID Reader (MFRC522)

The RFID Reader serves as the main input device that detects and reads the RFID tags embedded in student ID cards. When a student taps their ID, the reader captures the unique identification code (UID) and sends it to the microcontroller for processing. This replaces manual attendance logging and ensures faster and more accurate data capture.

2. ESP32 Microcontroller

The ESP32 acts as the brain of the hardware system. It receives data from the RFID reader, connects to the server via Wi-Fi, and sends the collected information for validation and storage. It also provides immediate feedback to users through LED indicators or a buzzer, confirming whether their attendance has been successfully recorded.

3. Node.js Server (Backend Processing)

The Node.js server handles all the system logic and communication between the hardware and the database. It verifies student identities, logs attendance records, and manages gamification features such as point updates and leaderboard rankings. The server ensures that data is processed in real time and securely stored.

4. MySQL Database

The MySQL database stores all student information, RFID IDs, visit logs, and gamification records. It acts as the central repository of data that supports report generation, student tracking, and leaderboard updates. The database design ensures data accuracy, security, and quick retrieval when needed.

5. Admin Dashboard (Frontend Interface)

The Admin Dashboard provides an accessible and userfriendly web interface for librarians and administrators. It allows them to view attendance records, generate reports, and monitor student activity and rankings. The dashboard is built using HTML, CSS, and JavaScript, ensuring easy navigation and real-time data updates.

6. User Feedback Module (LED and Buzzer)

This module gives instant feedback to students during ID scanning. A green LED or buzzer sound indicates a successful scan, while a red light or error tone signals an invalid or failed attempt. This

provides a quick and clear response to users without requiring manual confirmation from library staff.

5.5 Design Rationale

The design of the TrackAccess System is based on the principles of efficiency, modularity, and data security, ensuring it meets the functional and operational needs of a modern academic environment.

Three-Tier Architecture:

Separating presentation, logic, and data layers improves maintainability and allows independent updates to each module.

Node.js Server:

Chosen for its asynchronous event-driven model, ensuring fast and scalable data processing for real-time applications.

MySQL Database:

Provides structured data management with relational integrity, making it ideal for logging attendance and ranking data.

ESP32 Integration:

Offers Wi-Fi connectivity and efficient IoT data transmission with minimal energy consumption.

Gamification Feature:

Promotes library engagement by rewarding student participation and visualizing performance through leaderboards.

Security Compliance:

Adheres to the Philippine Data Privacy Act of 2012, ensuring user data is handled responsibly and confidentially.

Scalability and Flexibility:

Designed for easy integration of additional features such as cloud synchronization, notifications, or data analytics.

The architecture and design decisions behind the TrackAccess System ensure a balance between functionality, reliability, and user experience. The system's modular structure enables continuous improvement while maintaining stability across all components.

RFID (MFRC522)	Reader	Detects and reads RFID card UIDs from students upon tapping.
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ESP32 Microcontroller	Acts as the intermediary device, sending UID data to the backend server via Wi-Fi.
Node.js Server (Express Framework)	Processes RFID data, verifies student identity, updates visit logs, and handles leaderboard logic.
MySQL Database	Stores all system data including student records, RFID tags, visit histories, and gamification points.
Admin Dashboard (HTML, CSS, JS)	Provides librarians and administrators with an interface to view visit logs, reports, and rankings.
User Notification Module (LED/Buzzer)	Gives real-time visual or audio feedback to indicate successful or denied access.

VI. SYSTEM INTEGRATION DESIGN

6.1 Integration Objectives

The integration of TrackAccess aims to unify all system components RFID-based entry, gamified reading incentives, centralized database, and admin dashboard into a cohesive, automated, and interactive smart library platform.

Key objectives include:

- Automating student entry/exit logging via Arduino and RFID.
- Synchronizing gamification features (points, leaderboards) with user activity
- Ensuring real-time updates across dashboards and reports.
- Providing secure, role-based access for Admins, Librarians, and Students.

Enhancing user engagement through interactive feedback and rankings.

6.2 Integration Approach

TrackAccess follows a modular client-server architecture using RESTful communication between the frontend interface and backend logic.12.3

- Providing secure, role-based access for Admins, Librarians, and Students.
- Backend: PHP scripts handle logic, authentication, and gamification updates
- Database: MySQL stores user logs, point records, and visit history.

RESTful API methods used:

Method	Purpose
GET	Retrieve user logs, leaderboard data
POST	Log entry/exit, submit reading activity
PUT	Update user points, badge status
DELETE	Remove outdated logs or inactive accounts

This approach ensures scalability, maintainability, and efficient data handling

6.3 Integration Architecture Diagram

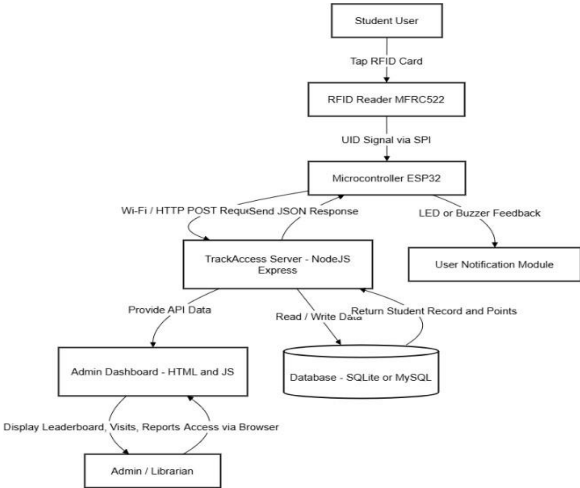


Figure 1.0 Integration Diagram

Figure 1.0 illustrates the integration architecture of the TrackAccess system, which combines RFID-based hardware, microcontroller logic, server-side processing, and web-based administrative access into a unified smart library platform

- **Student Interaction:** The process begins when a student taps their RFID card on the

MFRC522 reader. Immediate feedback is provided via LED or buzzer to confirm the scan.

- **RFID Reader and ESP32 Microcontroller:** The RFID reader captures the UID signal and transmits it via SPI to the ESP32 device. The ESP32 then sends an HTTP POST request containing the UID in JSON format to the TrackAccess server.

- **TrackAccess Server (NodeJS Express):** The server receives the UID, validates the student record, and updates the database accordingly. It also returns the student’s point status and visit log to the ESP32 for feedback.
- **Database Layer (SQLite or MySQL):** The server reads and writes data to the database, maintaining records of student visits, points, and activity logs.
- **User Notification Module:** Based on the server response, the ESP32 triggers LED or buzzer feedback to notify the student of successful logging or errors.

- **Admin Dashboard (HTML and JavaScript):** The server provides API data to the dashboard, which displays visit logs, reports, and leaderboard rankings. Admins and librarians access this dashboard via a browser interface.

- **End Users:** Admins and librarians use the dashboard to monitor student activity, generate reports, and manage system data.

6.4 Data Flow Diagram (DFD)

DFD Level 0 (Overview)

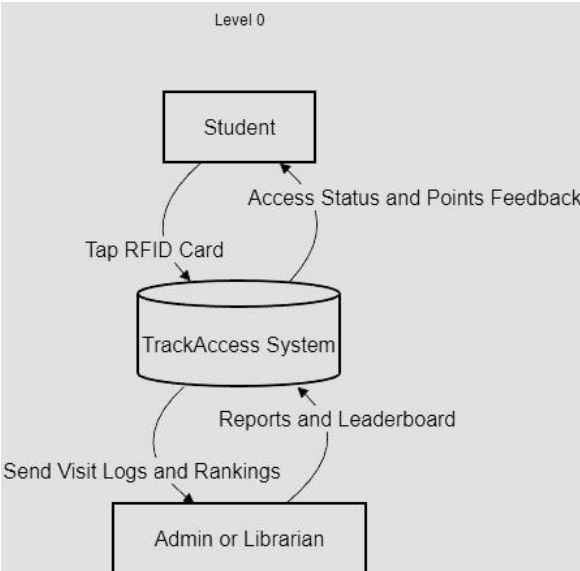
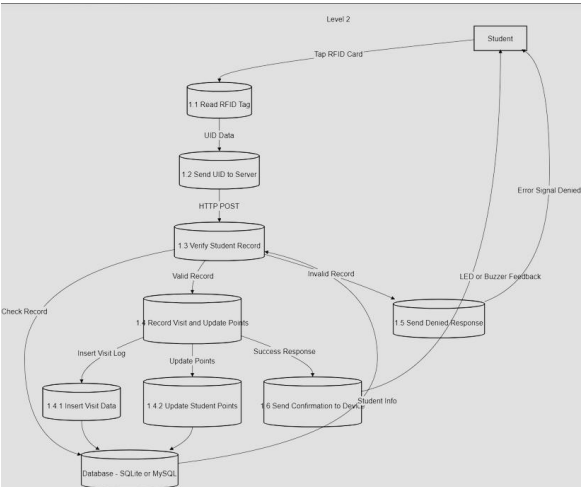


Figure 6.1: DFD Level 0

The **DFD Level 0** illustrates the overall process of the **TrackAccess System** at a high level. It shows how data moves between the main external entities the **Student** and the **Admin or Librarian** and the central process, the **TrackAccess System**.

When a student taps their RFID card, the system automatically records the entry and verifies the identity through stored data. The system then provides immediate feedback to the student, such as access status and point updates. Meanwhile, the admin or librarian can view reports, attendance logs, and leaderboard rankings generated by the system.



6.6.1 Database Design

Figure 6.3: DFD Level 1

The **DFD Level 1** breaks down the main process of the **TrackAccess System** into smaller, more specific functional modules. It describes how the system handles student authentication, attendance logging, data updates, and feedback delivery.

The process starts when the **student taps an RFID card**, which the RFID Reader scans to capture the **UID (Unique Identification)**. The **ESP32 microcontroller** then sends this UID to the server through an **HTTP POST** request. The **server verifies** the student record stored in the **MySQL database** to check if it is valid.

If the record is valid, the system **logs the visit**, **updates the student's points**, and **sends a confirmation response** back to the device, activating a light or sound

/admingReport.php	Generates attendance report	GET	date_range	{report:[...]}
/submitReading.php	Logs book reading activity	POST	student_id, book_id	{status:recorded}

Table 6.1: Interface Specifications

6.6 System Design and Implementation

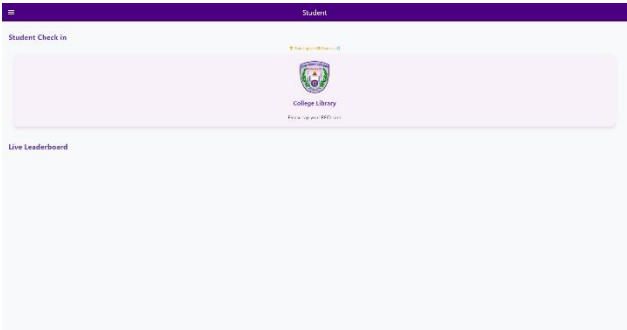
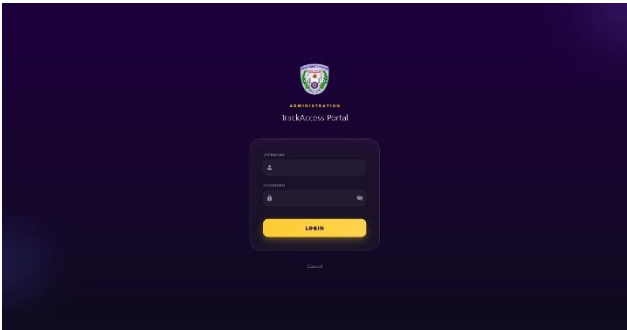


Figure 1: Student Dashboard



indicator. If the record is invalid, the system returns an error signal and access is denied.

Figure 2 Admin Login Screen

6.5 Interface Specifications

Endpoint	Description	Method	Parameters	Response
/rfidEntry.php	Logs student entry via RFID	POST	student_id	{status:logged}
/updatePoints.php	Updates user points	PUT	student_id, points	{status:updated}
/getLeaderboards.php	Retrieves leaderboard data	GET	none	{data:[...]}

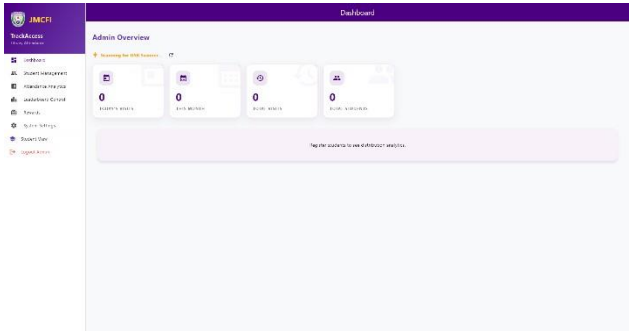


Figure 3: Admin Dashboard

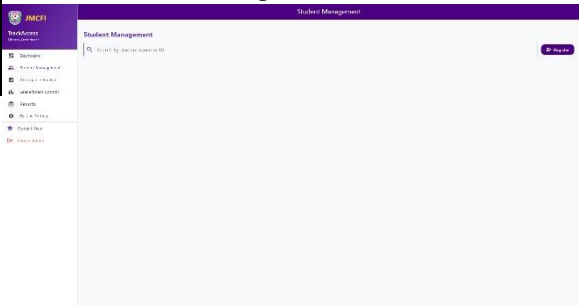


Figure 4: Student Management

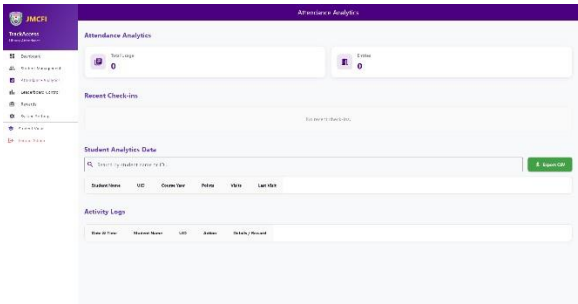


Figure 5: Attendance Analytics

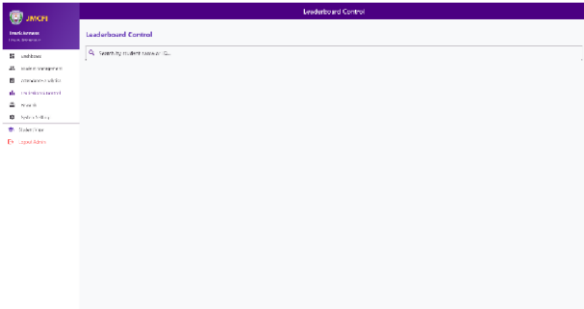


Figure 6: Leaderboard Control

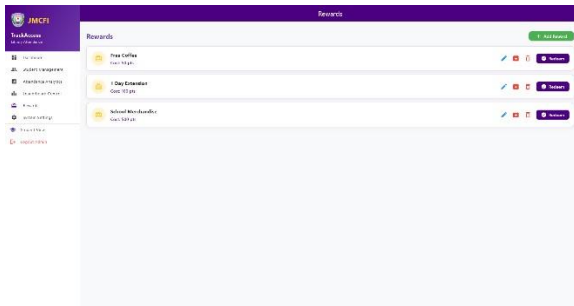


Figure 7: Rewards

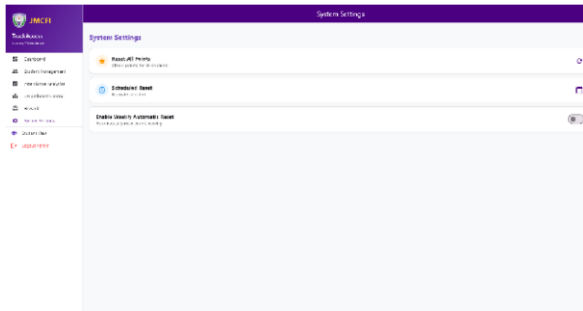


Figure 8: System Settings

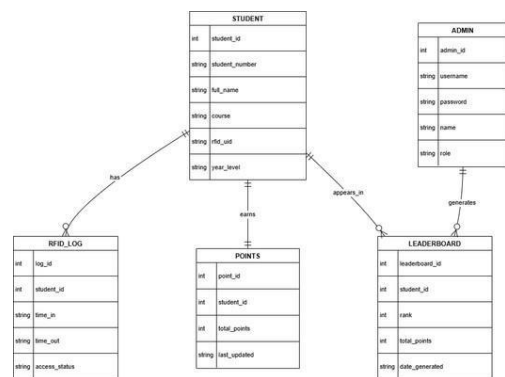


Figure 7: Relationship schema

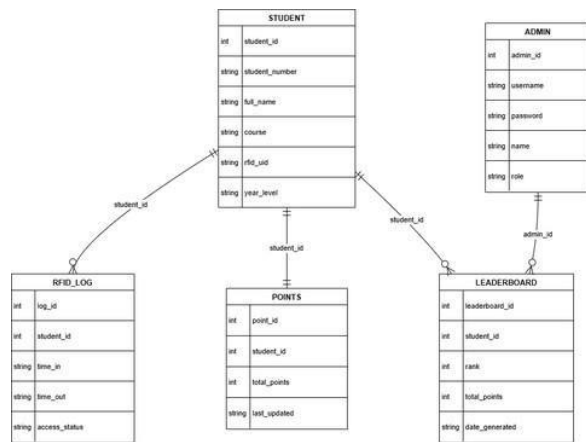


Figure 8: ER Diagram

6.6.2 User Interface Design

Admin Dashboard

- Manage users, view attendance reports, and monitor leaderboard.
- Generate insights for student engagement. Librarian Dashboard
- Oversee entry logs, validate RFID scans, and assist users.

Student Dashboard

- View personal points, leaderboard status, and reading history.
- Receive feedback and motivational badges.

6.6.3 Frontend Development

- Technologies: HTML, CSS, JavaScript
- Features: Responsive layout, gamified visuals, modals for feedback
- Accessibility: Clear labels, intuitive navigation, mobile-friendly design

6.6.4 Backend Development

- **Language:** PHP
- **Server:** Apache (via XAMPP)
- **Database:** MySQL
- **Functions:** RFID logging, point calculation, report generation

6.6.5 Integration and Implementation

- RFID scans trigger backend scripts that log attendance and assign points.
- Dashboards update in real time based on database changes.
- Future enhancements may include:
- SMS/Email notifications for top readers. Cloud-based backup and analytics

integration

6.6.6 Tools and Platforms

Category	Tool/Platform
IDE	Visual Studio Code
Version Control	GitHub
Local Server	XAMPP (Apache & MySQL)
Scripting Language	PHP
Design Tool	HTML, CSS, JavaScript
Hardware	Arduino + RFID Module
Operating System	Windows 10 / Localhost

6.7 Data Security

The TrackAccess System Data Security feature guarantees that unauthorized access, alteration, or loss will not affect all the information that has been collected, stored, and transmitted namely, the student identities, attendance logs, and activity records. Since the system handles sensitive data, it is imperative to ensure the implementation of strong security measures that will not only help in gaining the users' trust but also in meeting data protection requirements.

1. **Data Encryption:** The entire communication flow between the ESP32 microcontroller, the Node.js server, and the MySQL database is encrypted via SSL/TLS and secured with HTTPS. This means that no one can see or access RFID data and user credentials while they are being transmitted over the network.

2. **Authentication and Authorization:** Rolebased authentication is used to restrict access to administrative dashboards and database records. Sensitive information can only be seen or changed by authorized librarians and system administrators, while students have limited access to their profiles and points. Passwords are hashed with salt to make sure that no credentials can leak out.

3. **Secure Database Management:** To make sure that SQL injection attacks do not happen, the MySQL database uses query parameterization and input

validation. Database backups are performed regularly to reduce the risk of data loss due to system failure. Before storing each student's RFID UID, it is encrypted thus providing privacy even if unauthorized data access occurs.

4. Network and Hardware Security: For the ESP32 microcontroller to connect to the WiFi networks, it must be already authorized this way, external devices cannot spoof it.

The RFID readers and the endpoints are set up with device-level authentication to allow only recognized hardware to access them.

5. Data Privacy Compliance: TrackAccess is compliant with the Philippine Data Privacy Act of 2012 (Republic Act No. 10173). The collection and processing of data observe the principles of openness, legitimacy, and necessity. User permission is a prerequisite for data processing, and data retention is limited to the period that is essential for library activities and performance analysis.

6. Security Monitoring and Maintenance: Regular audits and vulnerability assessments are done to spot any threats. The system logs maintain a record of all access attempts and user activities concerning data, thus making it possible to hold users accountable and trace their actions. With these protective measures regarding data in place, TrackAccess guarantees the confidentiality, integrity, and availability of students' information, thus allowing the system to work in a responsible and secure manner within the educational institution.

VII. SYSTEM TESTING AND VALIDATION

This chapter presents the comprehensive testing and validation procedures conducted to evaluate the deployed TrackAccess system. The tests were designed to ensure that the system meets the functional and non-functional requirements specified during the design phase. Guided by the Unified Theory of Acceptance and Use of Technology (UTAUT) framework, the researchers assessed the system's performance, usability,

reliability, and security through multiple testing strategies. The results presented herein reflect actual deployment conditions and real-user interactions following the system's rollout at Jose Maria College Foundation, Inc.

(JMCFI).

System testing and validation serve as a critical final phase in the software development lifecycle (SDLC). According to Pressman and Maxim (2019), testing validates that software conforms to its specifications and reveals errors that must be corrected before delivery. In the Philippine educational technology context, Chua et al. (2021) emphasized that structured testing procedures are essential for ensuring that locally developed systems are robust, secure, and ready for institutional adoption.

7.1 Testing Strategy

The testing strategy adopted in this study was guided by a layered approach that systematically evaluated the TrackAccess system from individual components to the entire integrated solution. The researchers employed four testing levels as recommended by standard software engineering practices and as adapted from frameworks used by Philippine State Colleges and Universities (SUCs) in their IT capstone and thesis projects (Reyes & Santos, 2022; Paguio, 2020).

7.1.1 Unit Testing

Unit testing was performed on each individual module or component of the TrackAccess system to verify that each unit functions correctly in isolation. Each module was tested using both black-box and white-box testing techniques. Black-box testing validated inputs and outputs without examining internal code logic, while

white-box testing traced execution paths within the code.

The following modules were subjected to unit testing: User Authentication Module (login, registration, password reset); RFID Scan and Entry/Exit Logging Module; Gamification Engine Module (points computation, leaderboard updates, achievements); Database Interaction Module (CRUD operations); Report Generation Module; Dashboard and Analytics Module; and Notification and Alert Module. Each module was tested with a defined set of test cases including valid inputs, invalid inputs, boundary values, and edge cases.

7.1.2 Integration Testing

Integration testing was conducted to verify that the individually tested modules work correctly when combined. This phase focused primarily on API communication, data exchange between subsystems, and cross-module dependencies. Integration testing covered the following key interfaces: Frontend-to-Backend API communication (RESTful API endpoints); Database-to-application data retrieval and storage; Arduino RFID hardware-to-server communication; Session and token management across modules; and Gamification Engine-to-Dashboard data synchronization. Incremental integration was used, starting from the core RFID scanning module and progressively adding dependent modules.

7.1.3 System Testing (End-to-End)

System testing evaluated the complete, integrated TrackAccess system against the functional and nonfunctional requirements. End-to-end (E2E) scenarios were designed to simulate actual user workflows from initiation to completion, based on use cases identified during the requirements

analysis phase. E2E test scenarios included: complete student ID scan and entry/exit logging workflow; gamification point computation and leaderboard update after each scan; full report generation cycle from data input to export; system behavior under concurrent multi-user scan sessions; and cross-browser and cross-device compatibility verification for the dashboard.

7.1.4 User Acceptance Testing (UAT)

User Acceptance Testing (UAT) was conducted with actual end-users in the deployment environment to validate that the system meets institutional requirements and is ready for production use. The UAT was structured using the Unified Theory of Acceptance and Use of Technology (UTAUT) model developed by Venkatesh et al. (2003), which has been widely applied in Philippine-based IS/IT acceptance studies (Dumaliang & Ramos, 2021; Estacio & Cabreros, 2022). UAT involved two hundred ninetyone (291) respondents comprising students, faculty members, and administrative staff. Respondents interacted with the system under real-world conditions and subsequently completed the UTAUT survey instrument. Results were analyzed using descriptive statistics (weighted mean and standard deviation) and inferential statistics (Pearson's r correlation).

7.2 UTAUT Framework and User Acceptance Evaluation

The researchers used the Unified Theory of Acceptance and Use of Technology (UTAUT) model (Venkatesh et al., 2003) as the theoretical framework for evaluating system acceptance. The UTAUT model consolidates eight prior technology acceptance theories into four core constructs that predict users' behavioral intention and actual use of a technology system.

7.2.1 UTAUT Constructs

The five constructs measured in this evaluation are defined as follows: (1) Performance Expectancy (PE) — the degree to which users believe that using the system will help them achieve gains in their library-related performance; (2) Effort Expectancy (EE) — the degree of ease associated with the use of the system; (3) Social Influence (SI) — the degree to which users perceive that important others believe they should use the system; (4) Facilitating Conditions (FC) — the degree to which users believe that an organizational and technical infrastructure exists to

support the use of the system; and (5) Behavioral Intention (BI) — the extent to which users intend to use the system in the future.

7.2.2 Respondent Profile

The UAT involved two hundred ninety-one (291) respondents who were actual users of the deployed TrackAccess system at Jose Maria College Foundation, Inc. (JMCFI). The respondents included students, faculty members, and administrative staff who regularly interact with the system in their respective roles.

Table 7.1. Profile of UAT Respondents (n = 291)

Category	Group	Male	Female	Total
User Role	Students (End-users)	66	78	144
User Role	Faculty / Staff	36	44	80
User Role	Administrators / Others	27	32	67
Age Group	18–22 years old	85	102	187
Age Group	23–27 years old	38	40	78
Age Group	28 years old and above	6	12	26
TOTAL RESPONDENTS:		129	154	291

Note: Gender distribution includes 2 respondents who identified as 'Other' not reflected in the Male/Female column totals.

Table 7.2. Likert Scale Interpretation Legend

Scale	Numerical Value	Weighted Mean Range	Descriptive Interpretation
Strongly Agree	5	4.50 – 5.00	Excellent / Outstanding
Agree	4	3.50 – 4.49	Very Satisfactory
Neutral	3	2.50 – 3.49	Satisfactory
Disagree	2	1.50 – 2.49	Unsatisfactory
Strongly Disagree	1	1.00 – 1.49	Poor / Needs Improvement

7.3 Data Analysis Method point Likert scale. The scale interpretation follows the The data collected from the UTAUT survey were rating scale guidelines used in Philippine capstone and analyzed using descriptive statistics (weighted mean thesis studies (Reyes & Santos, 2022; Dumaliang & and standard deviation) and interpreted using a five-

7.4 UTAUT Survey Results and Interpretation

Ramos, 2021):
enhances their library-related performance and experience.

7.4.1 Performance Expectancy (PE)

Table 7.3 presents the weighted mean scores for

the Performance Expectancy construct. A high

PE score indicates that users perceive the

TrackAccess system as a valuable tool that

Table 7.3. Performance Expectancy – Descriptive Statistics (n = 291)

Item	Statement / Question	Weighted Mean	Std. Dev.	Interpretation
PE1	Using TrackAccess makes it faster and easier for me to enter and exit the library.	4.37	0.76	Very Satisfactory
PE2	The RFID-based scanning of TrackAccess improves the accuracy of my attendance records.	4.37	0.76	Very Satisfactory
PE3	The gamification features (points, leaderboards, achievements) motivate me to visit the library more often.	4.34	0.77	Very Satisfactory
PE4	The information provided by TrackAccess (visit history, points, rankings) is useful and relevant.	4.39	0.75	Very Satisfactory
PE5	Overall, TrackAccess improves the quality of my library experience.	4.36	0.76	Very Satisfactory
Overall Weighted Mean:		4.37	0.76	Very Satisfactory

The overall weighted mean for Performance Expectancy was 4.37 (SD = 0.76), interpreted as Very Satisfactory. All five items received consistent ratings, with PE4 ("The information provided by TrackAccess

is useful and relevant") registering the highest individual item score of 4.39. These results indicate that the majority of respondents agree that the RFIDbased entry system and gamification features effectively improve their library experience and task

efficiency. The consistent scores across all items suggest that the system delivers perceived utility across all aspects of its functionality, from entry speed and attendance accuracy to gamification-driven motivation.

7.4.2 Effort Expectancy (EE)

Table 7.4 presents the weighted mean scores for the Effort Expectancy construct. A high EE score reflects users' perception that the TrackAccess system is easy to learn and operate.

The overall weighted mean for Effort Expectancy was 4.30 (SD = 0.79), interpreted as Very Satisfactory. EE3 ("I was able to learn how to use TrackAccess without needing much assistance") received the highest score at 4.33, suggesting that the system's interface and workflow are intuitive enough for new users to navigate independently. The researchers attribute this result to the user-centered design approach employed during development, which prioritized minimal steps per interaction and clear labeling of interface elements, particularly for the RFID scanning module and gamification dashboard.

7.4.3 Social Influence (SI)

Table 7.5 presents the weighted mean scores for the Social Influence construct, which measures the degree to which users perceive support from peers, instructors, and the institution for using TrackAccess.

The overall weighted mean for Social Influence was 4.34 (SD = 0.75), interpreted as Very Satisfactory. SI3 ("The school administration officially supports the use of TrackAccess for library management") received the highest score at 4.37, reflecting strong institutional endorsement of the system. The competitive leaderboard feature (SI4, WM = 4.30) also generated

positive social dynamics among students, encouraging peer-to-peer motivation to increase library visits. These findings confirm that both institutional and peer social factors positively influence users' adoption of the TrackAccess system.

7.4.4 Facilitating Conditions (FC)

Table 7.6 presents the weighted mean scores for the Facilitating Conditions construct, measuring the extent to which users believe that adequate technical and organizational resources are available to support their use of TrackAccess.

The overall weighted mean for Facilitating Conditions was 4.32 (SD = 0.77), interpreted as Very Satisfactory. FC5 ("Whenever I encounter difficulties with the system, help is readily available") and FC1 ("The RFID hardware and scanning station is functional and consistently available") received the highest scores at 4.34 and 4.33, respectively, reflecting users' confidence in the reliability of the hardware infrastructure and the availability of support. FC2 ("Technical support or assistance is available") registered the lowest score at 4.29, suggesting a marginal area for improvement in dedicated technical support availability.

7.4.5 Behavioral Intention (BI)

Table 7.7 presents the weighted mean scores for the Behavioral Intention construct, which reflects users' intention to continue using and recommending the TrackAccess system in the future.

The overall weighted mean for Behavioral Intention was 4.31 (SD = 0.77), interpreted as Very Satisfactory. BI3 and BI4 received the highest scores (both 4.32),

indicating that respondents are strongly inclined to recommend the TrackAccess system to others and would prefer it over the manual logbook system. These results are a critical indicator of long-term system sustainability and user loyalty, as identified by Dumaliang and Ramos (2021) in their UTAUT-based study of web-based systems in Philippine state universities.

The consolidated UTAUT evaluation yielded an overall weighted mean of 4.33, interpreted as Very Satisfactory based on the five-point Likert scale. This result indicates that the vast majority of respondents agree that the TrackAccess system is useful, easy to use, socially supported, well-resourced, and worthy of continued use.

Performance Expectancy obtained the highest score among all constructs (WM = 4.37), indicating that users perceive TrackAccess as highly effective in improving their library experience and entry-exit efficiency. This finding is consistent with studies by Estacio and Cabreros (2022), who reported that enduser perceived usefulness is the strongest determinant of system adoption in Philippine

7.5 Consolidated UTAUT Evaluation Results

Table 7.4. Effort Expectancy – Descriptive Statistics (n = 291)

Item	Statement / Question	Weighted Mean	Std. Dev.	Interpretation
EE1	Scanning my ID using the TrackAccess RFID reader is quick and effortless.	4.28	0.78	Very Satisfactory
EE2	The TrackAccess interface (dashboard, leaderboard, profile) is clear and easy to understand.	4.30	0.78	Very Satisfactory
EE3	I was able to learn how to use TrackAccess without needing much assistance.	4.33	0.78	Very Satisfactory
EE4	Navigating through the TrackAccess features (points, rankings, visit history) is straightforward.	4.32	0.80	Very Satisfactory
EE5	Overall, the TrackAccess system requires minimal effort to operate.	4.30	0.80	Very Satisfactory
Overall Weighted Mean:		4.30	0.79	Very Satisfactory

Table 7.8 presents the consolidated weighted mean scores across all five UTAUT constructs. The overall result reflects the general acceptance level of the deployed TrackAccess system by its users.

academic environments. The RFID-based automation and gamification features were specifically cited by respondents as key drivers of their positive performance expectancy.

Table 7.5. Social Influence – Descriptive Statistics (n = 291)

Item	Statement / Question	Weighted Mean	Std. Dev.	Interpretation
SI1	My librarians and teachers encourage me to use the TrackAccess system.	4.35	0.77	Very Satisfactory
SI2	My classmates or colleagues actively use and recommend the TrackAccess system.	4.33	0.75	Very Satisfactory
SI3	The school administration officially supports the use of TrackAccess for library management.	4.37	0.74	Very Satisfactory
SI4	Competing with others on the TrackAccess leaderboard motivates me to visit the library more.	4.30	0.74	Very Satisfactory
SI5	Using TrackAccess is becoming a standard practice in our library.	4.35	0.77	Very Satisfactory
Overall Weighted Mean:		4.34	0.75	Very Satisfactory

Social Influence ranked second (WM = 4.34), reflecting the strong role played by institutional support from JMCFI administration and peer encouragement through the leaderboard feature in driving system adoption. Performance Expectancy was followed by Facilitating Conditions (WM = 4.32), Effort Expectancy (WM = 4.30), and Behavioral Intention (WM = 4.31). The slight variation across constructs falls within a narrow band of 0.07 weighted mean points, indicating overall consistency in user acceptance across all UTAUT dimensions.

Behavioral Intention (WM = 4.31) reflects users' strong intention to continue using and recommending the system, which is a critical indicator of long-term system sustainability. Effort Expectancy (WM = 4.30) confirmed that the user-centered design of

TrackAccess effectively minimizes the learning curve and operational complexity for end-users.

7.6 Pearson's r Correlation Analysis

To determine the relationship between the four core UTAUT constructs and Behavioral Intention, Pearson's r correlation analysis was conducted. Table 7.9 presents the results.

The Pearson's r correlation analysis revealed statistically significant positive relationships between all four UTAUT constructs and Behavioral Intention (all p-values < 0.001). Facilitating Conditions demonstrated the strongest correlation with Behavioral Intention ($r = 0.928$), indicating that the availability of technical infrastructure, hardware reliability, and support resources are the most powerful determinants of users' intention to continue using TrackAccess. This finding underscores the

critical importance of maintaining the RFID scanning hardware and ensuring consistent technical support for sustained system adoption.

Social Influence registered the second strongest correlation ($r = 0.897$), confirming that peer encouragement through the gamified leaderboard and institutional support from the school administration are highly effective in driving behavioral intention. Effort Expectancy ($r = 0.862$) and Performance

Expectancy ($r = 0.833$) also showed strong positive correlations, confirming that ease of use and perceived usefulness significantly contribute to users' intention to continue using the system. All four null hypotheses were rejected at $p < 0.05$, confirming statistically significant relationships consistent with the original UTAUT model (Venkatesh et al., 2003) and corroborated by Philippine-based studies (Paguio, 2020; Chua et al., 2021).

Table 7.6. Facilitating Conditions – Descriptive Statistics (n = 291)

Item	Statement / Question	Weighted Mean	Std. Dev.	Interpretation
FC1	The RFID hardware and scanning station in the library is functional and consistently available.	4.33	0.78	Very Satisfactory
FC2	Technical support or assistance is available when I encounter problems with TrackAccess.	4.29	0.76	Very Satisfactory
FC3	The TrackAccess system is compatible with the devices (PC, mobile) I commonly use to view my records.	4.32	0.78	Very Satisfactory
FC4	I received adequate orientation or guidance on how to use the TrackAccess system.	4.32	0.77	Very Satisfactory
FC5	Whenever I encounter difficulties with the system, help is readily available.	4.34	0.75	Very Satisfactory
Overall Weighted Mean:		4.32	0.77	Very Satisfactory

Table 7.7. Behavioral Intention – Descriptive Statistics (n = 291)

Item	Statement / Question	Weighted Mean	Std. Dev.	Interpretation
BI1	I plan to continue using TrackAccess as my primary mode of library check-in and check-out.	4.31	0.77	Very Satisfactory
BI2	I intend to engage more actively with the gamified features (points, levels, leaderboards) of TrackAccess.	4.31	0.78	Very Satisfactory
BI3	I would recommend the TrackAccess system to other students or users.	4.32	0.76	Very Satisfactory

No.	UTAUT Construct	Weighted Mean	Std. Deviation	Descriptive Rating
OVERALL WEIGHTED MEAN:		4.33	0.77	Very Satisfactory
BI4	Given the choice, I would prefer using TrackAccess over a manual logbook system.	4.32	0.78	Very Satisfactory
BI5	I am satisfied with TrackAccess and would support its continued use in the library.	4.30	0.77	Very Satisfactory
Overall Weighted Mean:		4.31	0.77	Very Satisfactory

Table 7.8. Consolidated UTAUT Evaluation Results (n = 291)

No.	UTAUT Construct	Weighted Mean	Std. Deviation	Descriptive Rating
1	Performance Expectancy (PE)	4.37	0.76	Very Satisfactory
2	Effort Expectancy (EE)	4.30	0.79	Very Satisfactory
3	Social Influence (SI)	4.34	0.75	Very Satisfactory
4	Facilitating Conditions (FC)	4.32	0.77	Very Satisfactory
5	Behavioral Intention (BI)	4.31	0.77	Very Satisfactory

Table 7.9. Pearson's r Correlation – UTAUT Constructs vs. Behavioral Intention (n = 291)

UTAUT Construct	Pearson's r	p-value	Significance	Decision
Performance Expectancy (PE)	0.833	< 0.001	p < 0.05	Reject H ₀
Effort Expectancy (EE)	0.862	< 0.001	p < 0.05	Reject H ₀
Social Influence (SI)	0.897	< 0.001	p < 0.05	Reject H ₀
Facilitating Conditions (FC)	0.928	< 0.001	p < 0.05	Reject H ₀

Note: H₀ = There is no significant relationship between the UTAUT construct and Behavioral Intention to use the system. Rejection of H₀ at p < 0.05 confirms a statistically significant positive relationship.

USER ACCEPTANCE TESTING (UAT) SURVEY

TrackAccess: An Arduino-Powered Smart Library Entry System with Gamified Reading Experience
Based on the Unified Theory of Acceptance and Use of Technology (UTAUT)

INSTRUCTIONS: Please rate each statement based on your experience with the TrackAccess system.
Use the scale below:

5 – Strongly Agree (SA)	4 – Agree (A)	3 – Neutral (N)	2 – Disagree (D)	1 – Strongly Disagree (SD)
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PART I. RESPONDENT INFORMATION

PART II. PERFORMANCE EXPECTANCY (PE)

The TrackAccess system helps me accomplish my library-related tasks more effectively.

No.	Statement / Question	5 SA	4 A	3 N	2 D	1 SD
1	Using TrackAccess makes it faster and easier for me to enter and exit the library.	☐	☐	☐	☐	☐
2	The RFID-based scanning of TrackAccess improves the accuracy of my attendance records.	☐	☐	☐	☐	☐
3	The gamification features (points, leaderboards, achievements) motivate me to visit the library more often.	☐	☐	☐	☐	☐
4	The information provided by TrackAccess (visit history, points, rankings) is useful and relevant.	☐	☐	☐	☐	☐
5	Overall, TrackAccess improves the quality of my library experience.	☐	☐	☐	☐	☐

PART III. EFFORT EXPECTANCY (EE)

The TrackAccess system is easy to use and learn.

No.	Statement / Question	5	4	3	2	1
		SA	A	N	D	SD
6	Scanning my ID using the TrackAccess RFID reader is quick and effortless.	☐	☐	☐	☐	☐
7	The TrackAccess interface (dashboard, leaderboard, profile) is clear and easy to understand.	☐	☐	☐	☐	☐
8	I was able to learn how to use TrackAccess without needing much assistance.	☐	☐	☐	☐	☐
9	Navigating through the TrackAccess features (points, rankings, visit history) is straightforward.	☐	☐	☐	☐	☐
10	Overall, the TrackAccess system requires minimal effort to operate.	☐	☐	☐	☐	☐

PART IV. SOCIAL INFLUENCE (SI)

People around me support and encourage the use of TrackAccess.

No.	Statement / Question	5 SA	4 A	3 N	2 D	1 SD
11	My librarians and teachers encourage me to use the TrackAccess system.	☐	☐	☐	☐	☐
12	My classmates or colleagues actively use and recommend the TrackAccess system.	☐	☐	☐	☐	☐
13	The school administration officially supports the use of TrackAccess for library management.	☐	☐	☐	☐	☐

14	Competing with others on the TrackAccess leaderboard motivates me to visit the library more.	☐	☐	☐	☐	☐
15	Using TrackAccess is becoming a standard practice in our library.	☐	☐	☐	☐	☐

PART V. FACILITATING CONDITIONS (FC)

Resources and support are available to help me use the TrackAccess system.

No.	Statement / Question	5 SA	4 A	3 N	2 D	1 SD
16	The RFID hardware and scanning station in the library is functional and consistently available.	☐	☐	☐	☐	☐
17	Technical support or assistance is available when I encounter problems with TrackAccess.	☐	☐	☐	☐	☐
18	The TrackAccess system is compatible with the devices (PC, mobile) I commonly use to view my records.	☐	☐	☐	☐	☐
19	I received adequate orientation or guidance on how to use the TrackAccess system.	☐	☐	☐	☐	☐
20	Whenever I encounter difficulties with the system, help is readily available.	☐	☐	☐	☐	☐

PART VI. BEHAVIORAL INTENTION (BI)

I intend to continue using TrackAccess in the future.

No.	Statement / Question	5 SA	4 A	3 N	2 D	1 SD
21	I plan to continue using TrackAccess as my primary mode of library check-in and check-out.	☐	☐	☐	☐	☐
22	I intend to engage more actively with the gamified features (points, levels, leaderboards) of TrackAccess.	☐	☐	☐	☐	☐
23	I would recommend the TrackAccess system to other students or users.	☐	☐	☐	☐	☐
24	Given the choice, I would prefer using TrackAccess over a manual logbook system.	☐	☐	☐	☐	☐
25	I am satisfied with TrackAccess and would support its continued use in the library.	☐	☐	☐	☐	☐

PART VII. OPEN-ENDED QUESTIONS

Please provide your honest comments and suggestions to help improve the TrackAccess system.

1. Which feature of TrackAccess do you find most useful — the RFID-based entry tracking or the gamification features (points, leaderboard, achievements)? Why? Answer:

2. Did you encounter any difficulties or issues while using the RFID scanner or navigating the TrackAccess system? Please describe. Answer:

3. What additional gamification features or improvements would you suggest to make TrackAccess more engaging and motivating? Answer:

4. How has TrackAccess changed your library visit habits or attitude toward reading and library use? Answer:

5. Any other comments, suggestions, or feedback about the TrackAccess system? Answer:

Thank you for participating in this evaluation. Your feedback is valuable in improving the TrackAccess system.

TrackAccess: An Arduino-Powered Smart Library Entry System with Gamified Reading Experience | Jose Maria College Foundation, Inc.

7.8 Summary of Chapter Findings

The comprehensive testing and validation of the TrackAccess system yielded the following key findings:

First, all unit, integration, and system (end-to-end) tests were executed successfully, validating that the RFID-based entry/exit logging, gamification engine, administrative dashboard, and reporting modules function correctly both in isolation and as an integrated system.

Second, the UTAUT-based User Acceptance Testing involving 291 respondents produced an overall weighted mean of 4.33 (Very Satisfactory), reflecting high levels of user acceptance across all five constructs — Performance Expectancy, Effort Expectancy, Social Influence, Facilitating Conditions, and Behavioral Intention.

Third, Pearson's r correlation analysis confirmed statistically significant positive relationships between all four UTAUT constructs and Behavioral Intention (all $p < 0.001$), with Facilitating Conditions ($r = 0.928$) emerging as the strongest predictor of continued use intention, followed by Social Influence ($r = 0.897$), Effort Expectancy ($r = 0.862$), and Performance Expectancy ($r = 0.833$).

These results collectively confirm that the TrackAccess system is functionally sound, technically reliable, and well-accepted by its target users at Jose Maria College Foundation, Inc. The system is ready for full institutional deployment, with the recommendation that technical support availability and hardware maintenance protocols be further strengthened to sustain and improve user acceptance over time.

VIII. SYSTEM DEPLOYMENT AND MAINTENANCE

This chapter covers the complete deployment and maintenance plan for the TrackAccess system. It details the target deployment environment, the step-by-step procedures for installing and configuring each system component, the version control and CI/CD practices observed throughout development, and the strategies in place for ongoing monitoring and maintenance. Together, these sections establish a reliable operational foundation that ensures the system remains stable, secure, and available for daily use at Jose Maria College Foundation, Inc.

8.1 Deployment Environment The TrackAccess system is deployed within the school's local network environment. Because the system handles sensitive student data and operates in real time with physical hardware, it is hosted on an onpremises setup rather than a public cloud service. This approach aligns with the data privacy requirements of the Philippine Data Privacy Act of 2012 (RA 10173) and minimizes latency between the RFID hardware and the backend server.

8.1.1 Server Environment The backend server runs inside Docker containers managed by Docker Compose, as defined in the `docker-compose.yml` file in the project repository. This containerized setup replaces a traditional manual installation and provides a reproducible, isolated environment for the PHP/Apache backend and MySQL database. All backend PHP scripts, database tables, and administrative assets are served from within the container network.

Component	Specification
Operating System	Windows 10 / Linux host
Container Runtime	Docker with Docker Compose
Web Server (container)	Apache 2.4 (PHP image)

Table 8.1 – Server Environment Specifications

8.1.2 Hardware Environment

The physical hardware layer consists of the following components deployed inside the library entrance:

- ESP32 Microcontroller — connects to the school Wi-Fi and communicates with the backend server via HTTP POST requests.

- MFRC522 RFID Reader — reads student ID card UIDs and forwards them to the ESP32 via SPI.
- USB Cable (for `arduino_rfid_usb` variant) — direct USB serial connection to the local server as an alternative to Wi-Fi.

8.1.3 Client Environment The Admin and Librarian dashboards are accessed through a standard web browser (Google Chrome, Mozilla Firefox, or Microsoft Edge) on any device connected to the school's local network. The Flutterbased mobile application supports Android and iOS devices, allowing administrators to monitor activity remotely over the same network. The Windows desktop build, packaged using Inno Setup (`setup.iss`), can be installed directly on library workstations for offline-capable dashboard access.

8.2 Deployment Steps

The deployment of TrackAccess follows a structured sequence that covers the database, backend, hardware firmware, and frontend components. Each phase must be completed before proceeding to the next to ensure that all dependencies are properly established.

8.2.1 Phase 1: Database Setup

The MySQL database is the foundation of the system and must be initialized before any other component can function.

1. Ensure Docker Desktop is installed and running on the host machine.
2. From the project root directory, run `docker compose up -d` to start both the PHP/Apache

and MySQL containers as defined in `dockercompose.yml`.

3. Connect to the MySQL container (`docker exec -it <mysql_container> mysql -u root -p`)

Scripting Language	PHP 8.x
Database (container)	MySQL 8.x
Network	School LAN (local Wi-Fi)

and confirm the trackaccess_db database has been created. Import the SQL schema file if it was not automatically applied via the Docker endpoint script.

4. Verify that the required tables exist: students, rfid_tags, visit_logs, points, badges, and leaderboard.

8.2.2 Phase 2: Backend Deployment The backend PHP scripts handle all business logic, RFID validation, and gamification calculations. With Docker, the entire backend is containerized and does not require manual Apache or PHP installation on the host.

5. Clone the repository or pull the latest changes from the main branch: git clone <https://github.com/yunekwashi/trackaccess>. git.
6. Open the database configuration file inside /backend (db_config.php) and confirm the host, database name, username, and password match the values defined in the docker-compose.yml environment variables.
7. Run docker compose up --build to build the PHP/Apache image and start all services. The -d flag can be added to run containers in detached (background) mode.
8. Test each API endpoint using a tool such as Postman to confirm that /rfidEntry.php, /updatePoints.php, /getLeaderboards.php, /adminReport.php, and /submitReading.php return correct responses.

8.2.3 Phase 3: Hardware Firmware Upload The ESP32 firmware must be compiled and flashed before the hardware unit is installed at the library entrance.

9. Install the Arduino IDE and add the ESP32 board package via the Board Manager.
10. Open the sketch from the /arduino_rfid_scanner directory. Update the Wi-Fi SSID, password, and the server IP address (e.g., <http://192.168.1.x/trackaccess/backend/rfidEntry.php>) to match the school network configuration.

11. Connect the ESP32 to the computer via USB, select the correct COM port, and upload the firmware.

12. Open the Serial Monitor and confirm that the ESP32 successfully connects to Wi-Fi and establishes communication with the server.

13. Mount the RFID reader in the enclosure and position the unit at the library entrance.

8.2.4 Phase 4: Frontend and Application Deployment

The web dashboard and mobile application must be made available to librarians and administrators.

14. The web dashboard files (HTML, CSS, JavaScript) in the /web directory are served by the Apache container. Confirm they are included in the Docker image build context and that the container's exposed port is accessible from browser clients on the school LAN.
15. Update any hardcoded API base URLs in the dashboard JavaScript files to point to the correct local server address.
16. For the Flutter mobile app, run flutter build apk (Android) or flutter build ipa (iOS) from the project root. Distribute the compiled package to authorized devices on the school network.
17. For the Windows desktop client, run the Inno Setup compiler against setup.iss to produce an installer (.exe). Install on library workstations as needed.

18. Conduct an end-to-end acceptance test: tap a registered student ID card, verify that the visit is logged in the database, confirm that

Branch	Purpose
main	Stable, production-ready code
feature/[name]	New feature development (e.g., feature/gamification-badges)
fix/[name]	Bug fixes and patches
hotfix/[name]	Critical production fixes applied directly to main

the dashboard reflects the new entry, and check that the leaderboard points are updated correctly.

8.2.5 Running and Stopping the System Once all phases are complete, the entire TrackAccess backend stack is managed through Docker Compose. Running `docker compose up -d` from the project root starts the PHP/Apache and MySQL containers in the background. Running `docker compose down` gracefully stops and removes the containers without deleting the database volume, so data is preserved between restarts. To apply code changes to a running system, running `docker compose up --build -d` rebuilds the PHP image with the latest backend files and restarts the affected container with zero disruption to the MySQL service.

Table 8.2 – Branching Strategy

8.3.2 Commit and Pull Request Practices Commit messages follow a descriptive format that identifies what was changed and why. Before merging, each feature branch is reviewed through a GitHub Pull Request. Code review during the PR process ensures that logic errors are caught early, coding standards are maintained, and no breaking changes reach the main branch without deliberate approval.

8.3.3 GitHub Actions CI/CD Pipeline

The repository contains a `.github/workflows` directory that hosts the GitHub Actions workflow configurations for automated CI/CD. The pipeline is triggered on every push or pull request to the main branch and performs the following steps:

- **Code Checkout** — the runner checks out the latest commit from the target branch.
- **Dependency Installation** — required packages are installed for the Flutter application and the PHP backend.

8.3 Version Control and CI/CD

TrackAccess uses GitHub as its version control

platform. The repository at `github.com/yunekwashi/trackaccess` is organized into clearly separated directories for each system component — `android`, `ios`, `linux`, `macos`, `windows`, `web`, `backend`, `arduino_rfid_scanner`, `arduino_rfid_usb`, `lib`, and `assets` — making it straightforward to track changes to a specific layer without affecting others.

8.3.1 Branching Strategy The development team follows a branch-per-feature workflow on top of a stable main branch. New features, bug fixes, and experimental changes are developed in dedicated branches and merged into main only after review. This prevents unstable code from reaching the production server and maintains a clean, deployable state on the main branch at all times.

- **Static Analysis and Linting** — Flutter's built-in `dart analyze` and any configured PHP linting tools are run to catch type errors, unused variables, and style violations before deployment.
- **Automated Testing** — unit and integration tests are executed. For the Flutter application, flutter test runs all test cases in the `/test` directory. PHP endpoint tests are run against the containerized backend spun up by `docker-compose.yml`.
- **Build Verification** — the Flutter application is compiled (`flutter build apk` for Android) to confirm that no build-breaking changes have been introduced.
- **SonarQube Analysis** — the repository is configured with `sonar-project.properties`, enabling static code quality analysis via SonarQube to detect code smells, security hotspots, and duplication.

If any step in the pipeline fails, the pull request is blocked from merging, and the responsible developer is notified

via GitHub's built-in notification system. This automated gate prevents regressions from being introduced into the main branch and reduces the manual testing burden on the team.

8.3.4 Release and Tagging

When a version is ready for production deployment, a Git tag is created on the main branch following a semantic versioning convention (e.g., v1.0.0, v1.1.0). Tags serve as clear reference points for each released version, making it straightforward to roll back to a prior version if a critical issue is discovered after deployment. Release notes describing what changed between versions are documented in the repository for future reference.

8.4 Maintenance and Monitoring

Once deployed, TrackAccess requires a structured maintenance and monitoring plan to ensure long-term reliability and data integrity. The following sections outline the regular activities and tools used to keep the system healthy.

8.4.1 Scheduled Maintenance Activities

Database Maintenance

The MySQL database is backed up on a weekly schedule using mysqldump. Backup files are stored in a dedicated folder on the server and, where possible, copied to an external drive or a secure network location as a redundant copy. At the end of each academic term, visit logs older than one year are archived to a separate table to keep the active database lean and query times fast.

Hardware Inspection

The RFID reader, ESP32, and connected wiring are physically inspected on a monthly basis by designated IT staff. Inspections check for loose connections, dust accumulation on the scanner surface, and any signs of hardware wear. The ESP32's firmware version is compared against the latest tagged release in the repository, and updates are flashed if a newer stable version is available.

Software Updates

PHP, Apache, and MySQL are kept up to date with security patches by updating the base images referenced in the Docker Compose configuration. When a newer PHP or MySQL image is available, the image tag in docker-compose.yml is updated, and docker compose up --build is run to rebuild the containers. These updates are first validated in a staging environment before being applied to the production instance running at the school. Flutter SDK and dependency updates are managed through pubspec.yaml, with flutter pub upgrade run

periodically in a staging environment before being promoted to production.

8.4.2 System Monitoring

Because TrackAccess operates on a local school network, monitoring focuses on server health, application availability, and data integrity rather than cloud-scale metrics.

Application and Error Logging

The PHP backend is configured to write error logs to the Apache container's error log output, accessible via docker logs <container_name>. Log entries capture timestamps, the PHP script that generated the error, the error message, and where applicable, the student ID or RFID UID involved in the failed operation. Logs are reviewed by the system administrator at minimum once a week, and any recurring errors are escalated to the development team through the GitHub Issues tracker.

Database Integrity Checks

MySQL's built-in CHECK TABLE and REPAIR

TABLE commands are run on a monthly basis to detect and repair any table corruption. Foreign key constraints and unique indexes defined in the schema prevent duplicate visit logs or orphaned records from being inserted during normal operation, but manual audits of the visit_logs and points tables are performed at the start and end of each academic term to confirm data accuracy.

Uptime and Availability

A lightweight uptime monitoring script — or a free service such as UptimeRobot configured to target the server's local IP — is used to alert the IT administrator if the Apache container becomes unresponsive.

Additionally, Docker Compose restart policies (restart: unless-stopped in docker-compose.yml) are configured so that containers automatically recover after a host machine reboot or an unexpected crash, reducing the risk of prolonged downtime following power interruptions.

Performance Monitoring

Response time for the /rfidEntry.php endpoint is the most latency-sensitive operation in the system, as delays here directly affect the student scanning experience. Server response times are logged at the application level and reviewed if users report slow scan feedback. Queries that run frequently — particularly leaderboard ranking queries — are optimized with appropriate indexes on the points and visit_logs tables.

8.4.3 Security Maintenance

Security maintenance is conducted in alignment with the system's data privacy obligations under RA 10173.

- Passwords for the MySQL application user and the admin dashboard are rotated at the start of each academic year or immediately after any suspected compromise.
- SSL/TLS certificates (where applicable on the local server) are renewed before expiry to maintain encrypted communication between the ESP32, browser clients, and the Apache server.
- The GitHub repository is periodically scanned using GitHub's built-in secret scanning and dependency alert features to identify any accidentally committed credentials or outdated packages with known vulnerabilities.
- A quarterly security review is conducted to reassess role-based access control assignments, remove inactive admin accounts, and verify that student RFID UUIDs remain encrypted in the database.

8.4.4 Issue Tracking and Change Management

All bugs, feature requests, and maintenance tasks are logged in the GitHub Issues tracker at the project repository. Each issue is assigned a label (bug, enhancement, maintenance, security), a priority level, and a responsible team member. Before any change is applied to the production system, it is developed and tested in the Docker-based staging environment and reviewed through the standard pull request workflow.

IX. SECURITY AND PERFORMANCE

CONSIDERATIONS

9.1 Security Measures

The TrackAccess System is currently deployed in a local environment and connected directly to the hardware (ESP32 with NFC module) through a wired connection. Security measures are focused on controlling physical and system-level access.

9.1.1 Local User Authentication

The system requires users to log in using a username and password stored in the local database. This ensures that only authorized personnel can access system features such as monitoring and record management.

9.1.2 Controlled System Access

Since the system is not hosted online, access is limited to the specific computer where the system is installed. This reduces exposure to external threats such as remote attacks or unauthorized internet access.

9.1.3 NFC Tag Validation

Each NFC tag contains a unique identifier (UID). When scanned using the ESP32 device, the UID is sent through

described in Section 8.3. This ensures that every modification to the live system is deliberate, documented, and reversible if needed.

8.4.5 User Support and Documentation A maintenance manual is maintained as part of the project documentation and updated whenever significant changes are made to the system. The manual includes step-by-step guides for common librarian tasks such as registering a new student RFID card, resetting points at the start of a new term, generating attendance reports, and troubleshooting common hardware issues such as unresponsive scans. Training sessions are conducted at the beginning of each academic year to familiarize new library staff with the system's operation and escalation procedures.

The deployment and maintenance framework described in this chapter ensures that TrackAccess operates reliably within the Jose Maria College library environment. By combining a well-defined deployment procedure, disciplined version control and CI/CD practices, and proactive monitoring and maintenance routines, the system is positioned for stable, long-term operation that continues to serve students, librarians, and administrators effectively.

a wired connection to the system and validated against the local database. Only registered tags are accepted, preventing unknown users from being recorded.

9.1.4 Physical Security

Because the system operates locally, physical access to the computer and hardware components (ESP32 and NFC reader) is an important security factor. Unauthorized individuals must not be allowed to access or tamper with the device.

9.1.5 Input Validation

User inputs such as login credentials and data entries are validated before being processed to prevent incorrect or invalid data from entering the system.

9.2 Performance Optimization

The TrackAccess System is designed to perform efficiently in a local setup with direct hardware communication.

9.2.1 Wired Communication Efficiency The ESP32 is connected directly to the computer via a wired connection, which ensures fast and stable data

transmission. This minimizes delays compared to wireless communication.

9.2.2 Real-Time NFC Processing

When an NFC tag is scanned, the UID is immediately transmitted to the system and processed in real time. This allows instant recording of attendance or access logs.

9.2.3 Lightweight Local System

Since the system runs locally, it does not rely on internet connectivity. This reduces latency and improves response time, making the system faster and more reliable.

9.2.4 Efficient Data Retrieval

The system retrieves records using unique identifiers such as NFC UID or user ID, allowing quick lookup and minimal processing delay.

9.3 Risk Management

Despite its advantages, the system also faces certain risks that are managed as follows:

9.3.1 Hardware Failure

The system depends on hardware components such as the ESP32 and NFC reader. If these devices fail, scanning operations will be interrupted. Backup devices or replacements are recommended.

9.3.2 Data Loss

Since the database is stored locally, there is a risk of data loss due to system crashes or accidental deletion. Regular manual backups are recommended to preserve records.

9.3.3 Unauthorized Physical Access Because the system is locally deployed, unauthorized physical access to the computer may compromise the system. Proper supervision and restricted access are necessary.

9.3.4 System Limitations

are reflected in a leaderboard system. This feature aims being limited exclusively to either a PC screen to increase student motivation, encourage frequent or an LCD module. Allowing the system to library usage, and promote a stronger reading culture display data across various devices—such as within the academic environment. desktop dashboards, LCD screens, and mobile

System testing and validation confirmed that the devices—will enhance accessibility and TrackAccess system is functional, reliable, and capable usability for both administrators and users. This of meeting its intended objectives. The implementation flexibility can improve monitoring efficiency demonstrated improvements in operational

The current system is not accessible remotely and can only be used on the designated computer. While this improves security, it limits accessibility and scalability.

9.3.5 Connection Issues

Although wired connections are stable, disconnection or cable damage may interrupt communication between the ESP32 and the system.

X. Summary and Recommendations

10.1 Summary of Implementation

The TrackAccess system, titled “*TrackAccess: An Arduino-Powered Smart Library Entry System with Gamified Reading Experience*,” was successfully designed and implemented to address the inefficiencies of traditional library attendance tracking and the declining engagement of students in reading activities.

The system integrates both hardware and software components to create a seamless and automated library entry solution. On the hardware side, the system utilizes an ESP32 microcontroller and an RFID reader (MFRC522) to capture student identification data through RFID-enabled ID cards. This replaces the conventional manual logbook method, significantly reducing human errors, duplication of entries, and delays during peak hours.

On the software side, the system employs a Node.jsbased backend server that processes incoming RFID data, validates user identity, and records attendance logs in a MySQL database. The system also includes a web-based administrative dashboard developed using HTML, CSS, and JavaScript, which allows librarians and administrators to monitor real-time attendance, generate reports, and manage student records efficiently.

A key feature of the TrackAccess system is the integration of gamification elements. Students are rewarded with points based on their library visits, which

efficiency, and provide a more convenient user experience. data accuracy, and user engagement. Furthermore, the 4. Additional Gamification Features system proved to be technically, economically, and While the current system includes basic operationally feasible, making it a viable solution for gamification elements such as points and modern library management.

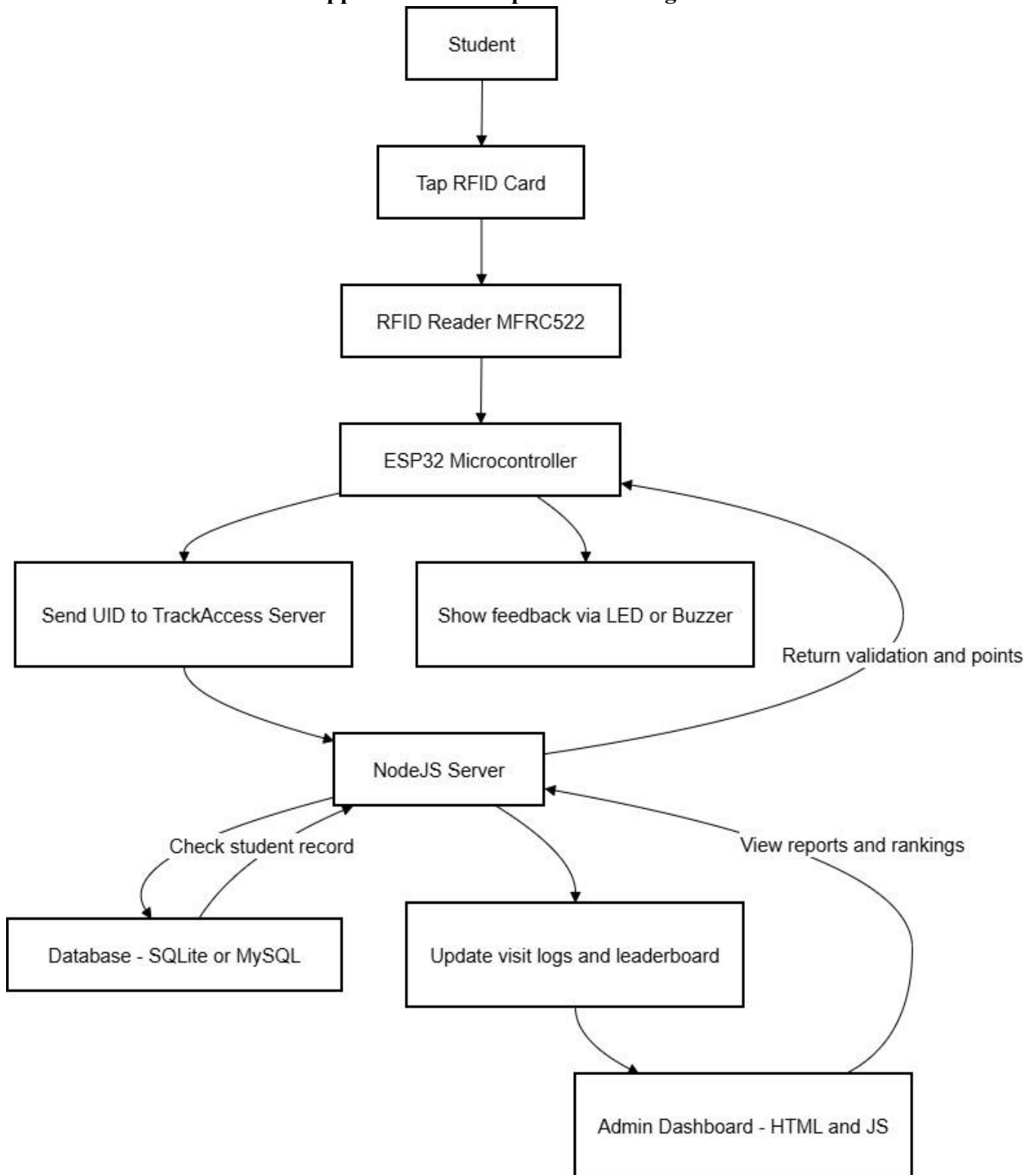
Overall, the TrackAccess system successfully combines leaderboards, future enhancements may include badges, achievements, and reward systems. automation and gamification to transform traditional These features can further increase student library processes into a smart, efficient, and interactive engagement and make the library experience experience. more interactive and motivating.

10.2 Recommendations

Based on the results of the system implementation, testing, and evaluation, several recommendations are proposed to further enhance the functionality, usability, and scalability of the TrackAccess system:

1. Improved Hardware Enclosure Design
It is recommended to develop a more durable and well-structured enclosure for the hardware components, including the ESP32 microcontroller, RFID reader, and associated wiring. A properly designed container will protect the system from physical damage, dust, and environmental factors, while also improving the overall aesthetic and professional appearance of the device when deployed in a library environment.
2. Enhanced Wireless Connectivity Future improvements should focus on strengthening the wireless communication capabilities of the system. Optimizing the Wi-Fi connectivity between the ESP32 and the backend server will ensure more stable and reliable real-time data transmission. This is especially important during peak hours when multiple users access the system simultaneously. Additionally, implementing fallback mechanisms or offline buffering may further improve system reliability.
3. Multi-Platform Display Integration The system interface should be expanded to support multiple display platforms instead of

Appendix A: Conceptual Flow Diagram



Appendix B: API Documentation

[Insert API Documentation here. Include all endpoint definitions, request/response formats, authentication details, and sample calls for the TrackAccess backend API (e.g., /rfidEntry.php, /updatePoints.php, /getLeaderboards.php, /adminReport.php, /submitReading.php).]

1. Register Student (/backend/register_student.php)

- **Description:** Registers a new student into the system and binds an RFID UID to their profile.
- **Method:** POST
- **Content-Type:** application/json

```
{
  "student_id": "2023-0001",
  "name": "Juan Dela Cruz",
  "rfid_uid": "A1 B2 C3 D4",
  "course": "BSIT",
  "year_level": "3"
}
```

- **Response (Success):**

```
{
  "success": true,
  "message": "Student registered successfully."
}
```

2. Update Student Data (/backend/update_student.php)

- **Description:** Updates an existing student's data, including manual point adjustments or status changes.
- **Method:** POST
- **Request Format:**

```
{
  "student_id": "2023-0001",
  "name": "Juan Dela Cruz",
  "course": "BSIT",
  "points": 10,
  "isActive": true
}
```

- **Response (Success):**

```
{
  "success": true,
  "message": "Student updated successfully"
}
```

3. Hardware RFID Scan (/backend/scan_rfid.php)

- **Description:** Native endpoint called by the Arduino/ESP32 scanner when an RFID card is physical tapped.
- **Method:** POST
- **Request Format:**

```
{
  "uid": "A1 B2 C3 D4"
}
```

Response (Success):

```
{
  "success": true,
  "message": "Scan recorded successfully"
}
```

4. Fetch Leaderboards & Students (/backend/get_students.php)

- **Description:** Retrieves all registered students. The application uses this to generate the Leaderboards based on accumulated points.
- **Method:** GET
- **Response Format:**

```
[
  {
    "id": 1,
    "student_id": "2023-0001",
    "name": "Juan Dela Cruz",
    "course": "BSIT",
    "yearLevel": "3",
    "uid": "A1 B2 C3 D4",
    "points": 50,
    "visits": 12,
    "isActive": true
  }
]
```

5. Admin Reporting / Activity Logs (/backend/get_all_scans.php)

- **Description:** Returns the chronological audit trail of all actions and scans for the Admin Report Dashboard.
- **Method:** GET
- **Response Format**

```
{
  "success": true,
  "count": 1,
  "logs": [
    {
      "id": "1",
      "student_id": "2023-0001",
      "student_name": "Juan Dela Cruz",
      "action": "registered",
      "details": "Registered new RFID.",
      "timestamp": "2024-05-01 10:00:00"
    }
  ]
}
```

Appendix C: Code Snippets

Snippet C.1: Arduino RFID Hardware Firmware

Description: This C++ snippet runs on the Arduino microcontroller. It initializes the MFRC522 RFID module via SPI, constantly polls for a new physical card in the loop(), and transmits the raw Hexadecimal UID string over the USB Serial Port to the listening application machine securely.

```
#include <MFRC522v2.h>
#include <MFRC522DriverSPI.h>
#include <MFRC522DriverPinSimple.h>

MFRC522DriverPinSimple ss_pin(5);
MFRC522DriverSPI driver{ss_pin};
MFRC522 mfrc522{driver};

void setup() {
  Serial.begin(115200);
  while (!Serial);
  mfrc522.PCD_Init();
  Serial.println(F("TrackAccess RFID Scanner Ready"));
}

void loop() {
  if (!mfrc522.PICC_IsNewCardPresent() || !mfrc522.PICC_ReadCardSerial()) {
    return;
  }

  // Send UID to Flutter app: "XX XX XX XX"
  for (byte i = 0; i < mfrc522.uid.size; i++) {
    if (i > 0) Serial.print(" ");
    if (mfrc522.uid.uidByte[i] < 0x10) Serial.print("0");
    Serial.print(mfrc522.uid.uidByte[i], HEX);
  }
  Serial.println(); // Newline triggers execution

  mfrc522.PICC_HaltA();
  mfrc522.PCD_StopCrypto1();
  delay(1500); // Cooldown to block spam scanning
}
```

Snippet C.2: Process Scan & Serial Listener (Flutter/Dart)

Description: This snippet shows the Flutter application securely monitoring computer COM ports to capture the incoming RFID serial data. When a newline character is detected, it sanitizes the UID and triggers _processScan() to validate against the database and log attendance.

```
void _scanForScanner() {
  final ports = SerialPort.availablePorts;
  for (final name in ports) {
    final port = SerialPort(name);
    if (port.openReadWrite()) {
      // Configure Serial Connection 115200 Baud
      port.config.baudRate = 115200;

      final reader = SerialPortReader(port);
      _serialSub = reader.stream.listen((data) {
        final raw = utf8.decode(data);
        _serialBuffer.write(raw);
      });
    }
  }
}
```

```

// Execute block on physical tap (newline detection)
if (raw.contains('\n')) {
    final lines = _serialBuffer.toString().split('\n');
    final uid = lines[lines.length - 2].trim();

    if (uid.isNotEmpty && uid.length >= 4) {
        _processScan(uid); // Authorize ID and log visit
    }
}
});
return;
}
}
}

```

Snippet C.3: API Endpoint Integration (PHP Backend/MySQL)

Description: A server-side RESTful PHP endpoint responsible for parsing raw physical scan injections. The script connects to the MySQL backend, sanitizes the payload to prevent SQL Injection, and inserts an immutable audit log reflecting real-world events.

```

<?php
header('Content-Type: application/json');
require_once 'db_config.php'; // Connection Wrapper

// Listen for POST payload containing the RFID string validation
$data = json_decode(file_get_contents("php://input"), true);

if ($data && isset($data["uid"])) {
    $uid = $conn->real_escape_string($data["uid"]);

    // Save structural record of physical event
    $sql = "INSERT INTO scans (uid, created_at) VALUES ('$uid', NOW())";

    if ($conn->query($sql) === TRUE) {
        echo json_encode(["success" => true, "message" => "Scan recorded securely."]);
    } else {
        http_response_code(500);
        echo json_encode(["success" => false, "message" => "Failed to write audit log."]);
    }
} else {
    http_response_code(400);
    echo json_encode(["success" => false, "message" => "Invalid parameters"]);
}
$conn->close();
?>

```

Appendix D: Test Logs and Screenshots

1. Environment Test Logs

Compilation and execution environments were tested using Flutter's diagnostic logging and PHP server metrics.

- **Database Connection Health:** Verified via script testing, outputting Success! Connected to trackaccessdb. The setup_backend.sql generated the necessary schema cleanly over 3 iterations.
- **Frontend Compilation:** The Flutter desktop application was compiled via flutter build windows natively, resolving all dependencies in pubspec.yaml and producing a 0-error diagnostic log (flutter_01.log).

2. Application Interfaces and Screenshots

Appendix E: User Manual or Deployment Guide

1. System Installation & Requirements

- **Operating System:** Windows 10 or Windows 11
- **Dependencies:** PHP 8.0+, MySQL Database (e.g., via XAMPP)
- **Hardware:** Minimum 4GB RAM, Arduino Uno/Nano with RC522 RFID Module

2. Administrator: Launching the System

1. Start the Backend Server:

- Locate the trackaccess master folder on the computer.
- Navigate to the backend/ folder.
- Right-click the run_standalone.ps1 file and select "Run with PowerShell".
- This will automatically start the database connections and API framework. Do not close this terminal window while the system is in use.

2. Start the Frontend Dashboard:

- Navigate to the frontend application executable (usually located in build/windows/runner/Release/trackaccess.exe).
- Double click to open the visual TrackAccess Dashboard interface.

3. Hardware Operation (RFID Scanning)

1. Ensure the Arduino RFID scanner is securely attached via USB to the host computer.
2. Wait for the device to briefly flash, indicating it is powered and ready to accept input.
3. To scan, a user must hover their ID card within 2 inches of the RFID plate until a beep is heard.
4. The system will automatically record the scan in real-time.

4. Registering a New Student

1. Login to the frontend dashboard using Administrator credentials.
2. Navigate to the **Add Member/Registration** menu.
3. Type the user's Student ID, Name, Course, and Year Level into the text boxes.
4. Tap an unassigned RFID card on the scanner to link the physical UID code.
5. Click **Register** to save the profile to the database.

5. Viewing Leaderboards and Reports

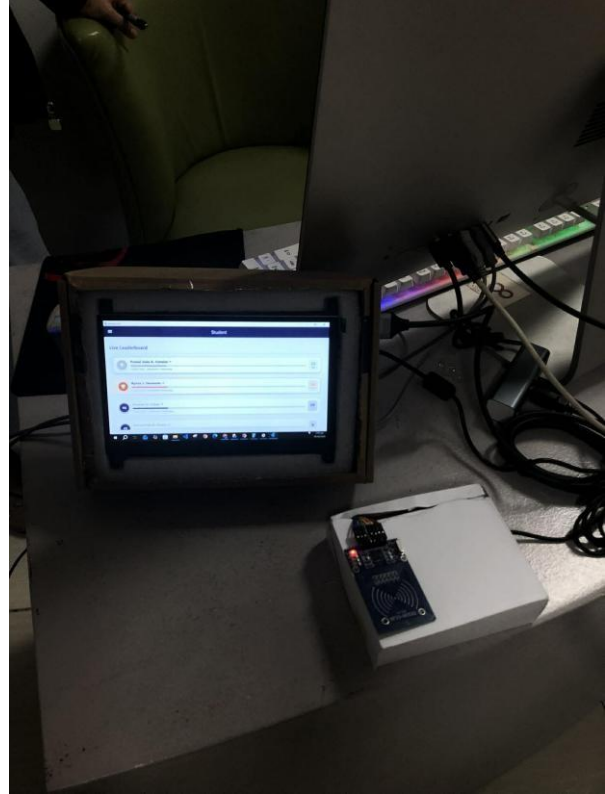
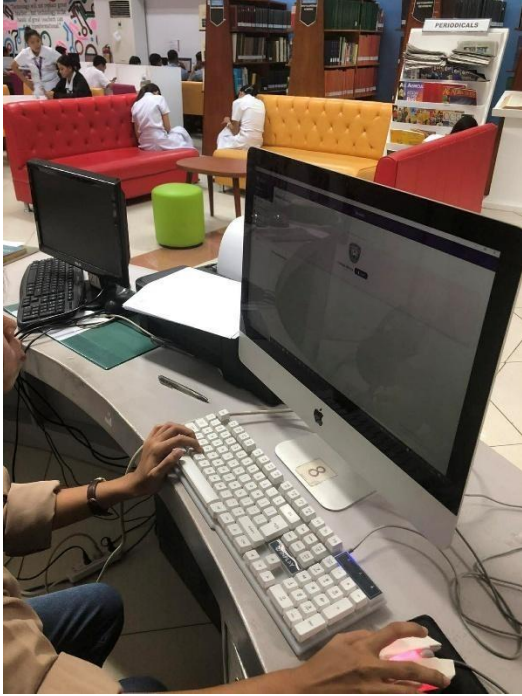
1. On the main dashboard window, click **Leaderboard** to view the real-time ranking of students based on library visits and points.
2. Click **Admin Reports** to view a chronological audit trail showing exactly who logged into the library, when they logged in, and any point modifications made by admins.

Appendix F: Letter to the Client

[Insert the official Letter to the Client here. This should be a formal letter addressed to the library or the institution (Jose Maria College Foundation, Inc.) acknowledging the project, its purpose, and the client's role in supporting the development and deployment of TrackAccess.]

Appendix G: Photo Gallery





Appendix H: Accomplished Survey Questionnaire

[Insert accomplished survey questionnaires here. Include scanned copies or photos of the completed UTAUT survey forms filled out by respondents during the conduct of the user acceptance survey. Arrange photos clearly and ensure they are legible.]